



Knowledge Based Systems

Laboratory activity

Name: Marcu Cristian Petric
Group: 30441
Email: marcupetric10@gmail.com

Teaching Assistant: Marius Adrian Stoica
Stoica.Ma.Marius@student.utcluj.ro



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Chapter 1

Determine Scope

Purpose: The ontology models a modern streaming platform ecosystem. It integrates production data (creators, release dates) with a robust content catalog (Films, Series, Episodes, Age Ratings) and consumer access data (streaming platforms, subscriptions, viewing history) to support intelligent content recommendation and access querying.

Competency Questions (SPARQL):

1. What are all the films in the catalog?
2. What are the names of all the subscribers?
3. Which streaming platforms are available in Romania?
4. What is the URL of each streaming platform?
5. Which subscription plans cost less than \$10?
6. Which artworks belong to the Science Fiction genre?
7. What plans does Netflix offer?
8. Which subscribers have watched artworks that their friends also liked?
9. Which subscribers have watched at least one science fiction film?
10. Which subscribers are subscribed to a platform that is not available in their country?
11. What are all reviews with their score and the name of the reviewed artwork?
12. Who are the five oldest subscribers?
13. Which subscribers liked an artwork but never wrote a review for it?
14. What are all the episodes of *Better Call Saul* Season 1?
15. For each subscriber, what plan are they on and what does it cost — including subscribers with no plan?
16. How many artworks has each subscriber watched, sorted from most to least?
17. What is the average review score per genre, for genres with at least two reviews?
18. Starting from the first episode of *Better Call Saul* Season 1, what is the full episode sequence?

19. Which pairs of subscribers are friends and have both watched the same artwork?
20. For each platform, what is the cheapest, most expensive, and average plan cost?
21. Which subscribers are older than 25?
22. Which reviews have a score of 7 or above?
23. Which subscription plans have no advertisements?
24. If a subscriber has watched a series, which seasons have they also implicitly watched?
25. Which artworks should be recommended to a subscriber based on what their friends like?
26. Which artworks should be recommended to a subscriber based on shared genre preferences?
27. Which subscribers have both a premium plan and are located in a country where their platform is available?
28. Which films should be recommended to a subscriber because a friend likes the same genre and the film has a score of at least 9?
29. Which pairs of subscribers are friends who have watched the same artwork?
30. For a subscriber who has watched at least one episode in a season, which other episodes in that season are considered in progress?

Chapter 2

Consider Reuse

For this project, I have chosen to extend an existing ontology rather than building one from scratch.

- **The ontology used:** The Film Ontology.
- **Source/Link:** <http://semantics.id/ns/example/film/>
- **Author(s):** Fajar J. Ekaputra (Contributors: Filip Kovacevic, Laura Waltersdorfer, Marta Sabou).
- **Extension Strategy:** The base ontology effectively models film creation (artists, studios). I will extend it by adding a “Consumer/Streaming” domain to model where these films and series are hosted, the episodic nature of content, robust content classification, and user viewing habits.

Base Ontology Description:

The original Film Ontology, developed for the Information Systems and Semantic Web course at TU Wien, models the production side of the film industry. The following classes and properties from the base ontology are reused directly:

- **Classes:** Artwork, Film, Soundtrack, Trailer, FilmStudio, Genre, Performer, Actor, Director, Composer, ScriptWriter, Producer, Cinematographer, Editor, CostumeDesigner, foaf:Person
- **Object properties:** hasActor, hasDirector, hasCrew, hasComposer, hasScriptWriter, hasPerformer, hasGenre, hasFilmStudio, hasSoundtrack, prequel, sequel, friendOf, foaf:knows
- **Data properties:** budget, dateOfBirth, duration, establishedDate, fullName, gender, releaseYear

Chapter 3

Enumerate Important Terms

The relevant terms for both the base ontology and the new extensions are listed below:

- **Nouns (candidate classes):**

- *From Base:* Film, Actor, Director, Soundtrack, Artwork, Person, Composer, FilmStudio, Performer, ScriptWriter, Producer, Cinematographer, Editor, CostumeDesigner, Genre, Trailer.
- *New Extensions:* Series, Miniseries, Season, Episode, AgeRating, Subscriber, StreamingPlatform, SubscriptionPlan, BasicPlan, PremiumPlan, FamilyPlan, AdSupportedPlan, StudentPlan, Country, Review, Award, Oscar, Emmy, GoldenGlobe, PalmeDOr, SciFiFilm, EnsembleFilm, AwardWinningArtwork, AdultSubscriber, PositiveReview, AdFreePlan, FullAccessSubscriber.

- **Verbs (candidate object properties):**

- *From Base:* hasActor, hasDirector, hasCrew, hasComposer, hasScriptWriter, hasPerformer, hasFilmStudio, hasSoundtrack, hasGenre, prequel, sequel, friendOf, knows.
- *New Extensions:* watched, likes, wroteReview, reviewsArtwork, subscribedTo, hasPlan, offersPlan, availableInCountry, availableOn, locatedIn, hasOriginCountry, hasAgeRating, hasAward, hasSeason, hasEpisode, nextEpisode, belongsToSeries, inSameUniverse, coStarredWith, recommendsTo, coWatched, inProgress.

- **Descriptive nouns (candidate data properties):**

- *From Base:* releaseYear, fullName, dateOfBirth, budget, duration, gender, establishedDate.
- *New Extensions:* subscriberAge, subscriptionCost, maxSimultaneousStreams, hasAds, platformURL, reviewScore, imdbRating, episodeNumber, seasonNumber, synopsis, originalLanguage, registrationDate, awardYear.

Chapter 4

Define Classes

The class hierarchy based on the base ontology and our streaming platform extensions, totaling 46 classes:

- **Artwork**
 - Film
 - * ShortFilm (*New*)
 - * SciFiFilm (*New, Defined*)
 - * EnsembleFilm (*New, Defined*)
 - Series (*New*)
 - * Miniseries (*New*)
 - Season (*New*)
 - Episode (*New*)
 - Soundtrack
 - Trailer (*New*)
 - AwardWinningArtwork (*New, Defined*)
- **Genre**
- **AgeRating** (*New*)
- **Person** (foaf:Person)
 - Director
 - Composer
 - ScriptWriter
 - Producer (*New*)
 - Cinematographer (*New*)
 - Editor (*New*)
 - CostumeDesigner (*New*)
 - Performer
 - * Actor
 - VoiceActor (*New*)

- Subscriber (*New*)
 - * AdultSubscriber (*New, Inferred*)
 - * FullAccessSubscriber (*New, Inferred*)
- **FilmStudio**
- **StreamingPlatform** (*New*)
- **SubscriptionPlan** (*New*)
 - BasicPlan (*New*)
 - PremiumPlan (*New*)
 - FamilyPlan (*New*)
 - AdSupportedPlan (*New*)
 - StudentPlan (*New*)
 - AdFreePlan (*New, Inferred*)
- **Country** (*New*)
- **Review** (*New*)
 - PositiveReview (*New, Inferred*)
- **Award** (*New*)
 - Oscar (*New*)
 - Emmy (*New*)
 - * BestComedy (*New*)
 - * BestDrama (*New*)
 - GoldenGlobe (*New*)
 - PalmeDOr (*New*)

Classes marked *Defined* use OWL restrictions and are automatically classified by the reasoner.
 Classes marked *Inferred* are populated by SWRL rules.

Chapter 5

Define Roles

Object properties (relations between classes) - Total: 37

- `hasActor` (Film $\rightarrow$ Actor) (*Base*)
- `hasDirector` (Film $\rightarrow$ Director) (*Base*)
- `hasCrew` (Film $\rightarrow$ Person) (*Base*)
- `hasComposer` (Film $\rightarrow$ Composer) (*Base*)
- `hasScriptWriter` (Film $\rightarrow$ ScriptWriter) (*Base*)
- `hasPerformer` (Artwork $\rightarrow$ Performer) (*Base*)
- `hasGenre` (Film $\rightarrow$ Genre) (*Base*)
- `hasFilmStudio` (Film $\rightarrow$ FilmStudio) (*Base*)
- `hasSoundtrack` (Film $\rightarrow$ Soundtrack) (*Base*)
- `prequel` (Film $\rightarrow$ Film) (*Base*)
- `sequel` (Film $\rightarrow$ Film) (*Base*)
- `friendOf` (Person $\rightarrow$ Person) (*Base*)
- `knows` (Person $\rightarrow$ Person) (*Base*)
- `watched` (Subscriber $\rightarrow$ Artwork) (*New*)
- `likes` (Subscriber $\rightarrow$ Artwork) (*New*)
- `availableOn` (Artwork $\rightarrow$ StreamingPlatform) (*New*)
- `availableInCountry` (StreamingPlatform $\rightarrow$ Country) (*New*)
- `offersPlan` (StreamingPlatform $\rightarrow$ SubscriptionPlan) (*New*)
- `hasOriginCountry` (Artwork $\rightarrow$ Country) (*New*)
- `locatedIn` (Subscriber $\rightarrow$ Country) (*New*)
- `belongsToSeries` (Episode $\rightarrow$ Series) (*New*)
- `hasSeason` (Series $\rightarrow$ Season) (*New*)

- hasEpisode (Season $\rightarrow$ Episode) (*New*)
- nextEpisode (Episode $\rightarrow$ Episode) (*New*)
- hasPlan (Subscriber $\rightarrow$ SubscriptionPlan) (*New*)
- subscribedTo (Subscriber $\rightarrow$ StreamingPlatform) (*New*)
- hasAgeRating (Artwork $\rightarrow$ AgeRating) (*New*)
- hasAward (Artwork $\rightarrow$ Award) (*New*)
- wroteReview (Subscriber $\rightarrow$ Review) (*New*)
- reviewsArtwork (Review $\rightarrow$ Artwork) (*New*)
- hasCinematographer (Film $\rightarrow$ Cinematographer) (*New*)
- hasEditor (Film $\rightarrow$ Editor) (*New*)
- hasProducer (Film $\rightarrow$ Producer) (*New*)
- hasCostumeDesigner (Film $\rightarrow$ CostumeDesigner) (*New*)
- coStarredWith (Actor $\rightarrow$ Actor) (*New*)
- inSameUniverse (Artwork $\rightarrow$ Artwork) (*New*)
- recommendsTo (Subscriber $\rightarrow$ Subscriber) (*New*)
- coWatched (Subscriber $\rightarrow$ Subscriber) (*New*)
- inProgress (Subscriber $\rightarrow$ Episode) (*New*)

Data properties (attributes) - Total: 20

- budget (Film $\rightarrow$ float) (*Base*)
- dateOfBirth (Person $\rightarrow$ date) (*Base*)
- duration (Artwork $\rightarrow$ float) (*Base*)
- establishedDate (FilmStudio $\rightarrow$ date) (*Base*)
- fullName (Person $\rightarrow$ string) (*Base*)
- gender (Person $\rightarrow$ string) (*Base*)
- releaseYear (Film $\rightarrow$ integer) (*Base*)
- subscriberAge (Subscriber $\rightarrow$ integer) (*New*)
- imdbRating (Artwork $\rightarrow$ decimal) (*New*)
- subscriptionCost (SubscriptionPlan $\rightarrow$ decimal) (*New*)
- episodeNumber (Episode $\rightarrow$ integer) (*New*)
- seasonNumber (Season $\rightarrow$ integer) (*New*)

- `synopsis` (`Artwork` $\rightarrow$ `string`) (*New*)
- `originalLanguage` (`Artwork` $\rightarrow$ `string`) (*New*)
- `maxSimultaneousStreams` (`SubscriptionPlan` $\rightarrow$ `integer`) (*New*)
- `platformURL` (`StreamingPlatform` $\rightarrow$ `string`) (*New*)
- `hasAds` (`SubscriptionPlan` $\rightarrow$ `boolean`) (*New*)
- `registrationDate` (`Subscriber` $\rightarrow$ `date`) (*New*)
- `awardYear` (`Award` $\rightarrow$ `integer`) (*New*)
- `reviewScore` (`Review` $\rightarrow$ `integer`) (*New*)

Annotation properties:

- `rdfs:label` (Used for human-readable naming)
- `dc:creator` (Used to track who created the ontology)
- `rdfs:comment` (Used for adding descriptions to new classes)

Stage 5 Extension: Advanced Role Characteristics

To fulfill the logical constraints of the ontology and map the complex relationships within a streaming platform ecosystem, we defined specific use-cases for the advanced OWL relation characteristics:

- **Functional Property:** `hasAgeRating` (`Artwork` $\rightarrow$ `AgeRating`). A specific piece of artwork can only hold one official age classification.
- **Inverse Functional Property:** `wroteReview` (`Subscriber` $\rightarrow$ `Review`). A specific review can uniquely identify its author; a single review cannot be authored by two distinct subscribers.
- **Transitive Property:** `sequel` (`Film` $\rightarrow$ `Film`). If Film C is a sequel to Film B, and Film B is a sequel to Film A, the reasoner infers that Film C is in the sequel lineage of Film A.
- **Symmetric Property:** `coStarredWith` (`Actor` $\rightarrow$ `Actor`). If Actor A co-starred with Actor B, then Actor B automatically co-starred with Actor A.
- **Asymmetric Property:** `nextEpisode` (`Episode` $\rightarrow$ `Episode`). If Episode 2 follows Episode 1, it is mathematically impossible for Episode 1 to follow Episode 2.
- **Reflexive Property:** `inSameUniverse` (`Artwork` $\rightarrow$ `Artwork`). By default, every piece of artwork exists within the same cinematic universe as itself.
- **Irreflexive Property:** `recommendsTo` (`Subscriber` $\rightarrow$ `Subscriber`). A subscriber can recommend content to other users on the platform, but cannot logically send a recommendation to themselves.

Chapter 6

Role Constraints

Stage 6a — OWL Restrictions (Defined Classes)

To enable automated reasoning, three classes are defined using OWL restrictions. These are *defined classes* — the reasoner automatically classifies any individual that satisfies the conditions, without requiring explicit assertion.

- **SciFiFilm** — Value restriction:
 - A **Film** that **hasGenre** exactly **genre_science_fiction**.
 - *Restriction type:* `owl:hasValue`
 - *DL notation:* $\text{SciFiFilm} \equiv \text{Film} \sqcap \exists \text{hasGenre}.\{\text{genre_science_fiction}\}$
 - *Example:* **movie_inception** is asserted as a **Film** with **hasGenre genre_science_fiction**. After running the reasoner, it is automatically classified as **SciFiFilm**.
- **EnsembleFilm** — Cardinality restriction:
 - A **Film** that **hasActor** at least 3 instances of **Actor**.
 - *Restriction type:* `owl:minQualifiedCardinality 3`
 - *DL notation:* $\text{EnsembleFilm} \equiv \text{Film} \sqcap (\geq 3 \text{ hasActor}.\text{Actor})$
 - *Example:* Any film with three or more actor assertions is automatically inferred as an **EnsembleFilm** by the reasoner.
- **AwardWinningArtwork** — Existential restriction:
 - An **Artwork** that **hasAward** some **Award**.
 - *Restriction type:* `owl:someValuesFrom`
 - *DL notation:* $\text{AwardWinningArtwork} \equiv \text{Artwork} \sqcap \exists \text{hasAward}.\text{Award}$
 - *Example:* Any artwork with at least one **hasAward** assertion is automatically classified as **AwardWinningArtwork**.

Stage 6b — SWRL Rules

Ten SWRL rules were defined to derive new knowledge from existing facts. Rules are executed using the Drools rule engine inside the SWRLTab in Protégé, and the inferred facts are subsequently transferred back to the OWL ontology for reasoning.

- **Rule 1 — AdultSubscriber:** Classifies subscribers older than 25.

```
Subscriber(?s) ^ subscriberAge(?s, ?age) ^ swrlb:greaterThan(?age, 25)
-> AdultSubscriber(?s)
```

- **Rule 2 — PositiveReview:** Classifies reviews with a score of 7 or above.

```
Review(?r) ^ reviewScore(?r, ?score) ^ swrlb:greaterThanOrEqual(?score, 7)
-> PositiveReview(?r)
```

- **Rule 3 — Watched season:** If a subscriber watched a series, they implicitly watched its seasons.

```
Subscriber(?s) ^ watched(?s, ?series) ^ Series(?series)
^ hasSeason(?series, ?season) -> watched(?s, ?season)
```

- **Rule 4 — AdFreePlan:** Classifies plans with no advertisements.

```
SubscriptionPlan(?p) ^ hasAds(?p, false) -> AdFreePlan(?p)
```

- **Rule 5 — recommendsTo (friend-based):** Recommends artworks liked by a subscriber's friends.

```
Subscriber(?a) ^ friendOf(?a, ?b) ^ likes(?b, ?x) -> recommendsTo(?x, ?a)
```

- **Rule 6 — recommendsTo (genre-based):** Recommends artworks of the same genre as something a subscriber already liked.

```
Subscriber(?s) ^ likes(?s, ?a1) ^ hasGenre(?a1, ?g)
^ hasGenre(?a2, ?g) -> recommendsTo(?a2, ?s)
```

- **Rule 7 — FullAccessSubscriber:** Classifies subscribers on a premium plan whose platform is available in their country.

```
Subscriber(?s) ^ hasPlan(?s, ?p) ^ PremiumPlan(?p)
^ subscribedTo(?s, ?platform) ^ offersPlan(?platform, ?p)
^ locatedIn(?s, ?c) ^ availableInCountry(?platform, ?c)
-> FullAccessSubscriber(?s)
```

- **Rule 8 — recommendsTo (high-rated genre match):** Recommends films in the same genre liked by a friend, with a review score of at least 9.

```
Subscriber(?s) ^ friendOf(?s, ?f) ^ likes(?f, ?a1) ^ hasGenre(?a1, ?g)
^ Film(?film) ^ hasGenre(?film, ?g) ^ Review(?r)
^ reviewsArtwork(?r, ?film) ^ reviewScore(?r, ?score)
^ swrlb:greaterThanOrEqual(?score, 9) -> recommendsTo(?film, ?s)
```

- **Rule 9 — coWatched:** Infers that two friends who watched the same artwork have co-watched it.

```
Subscriber(?a) ^ Subscriber(?b) ^ friendOf(?a, ?b)
^ watched(?a, ?x) ^ watched(?b, ?x) -> coWatched(?a, ?b)
```

- **Rule 10 — inProgress:** Marks episodes as in-progress for a subscriber who has watched another episode in the same season.

```
Subscriber(?s) ^ watched(?s, ?e1) ^ Episode(?e1)
^ hasEpisode(?season, ?e1) ^ hasEpisode(?season, ?e2)
^ Episode(?e2) -> inProgress(?s, ?e2)
```

Chapter 7

Create Instances

Individuals were populated using **Cellfie**, a Protégé plugin that transforms spreadsheet data into OWL axioms via declarative mapping rules. Three spreadsheets were used, each with its own set of rules exported as JSON for reproducibility.

The ontology was populated with a total of **152 named individuals** across the following classes:

Class	Count
Film	13
Series	4
Miniseries	1
Season	4
Episode	34
Genre	7
StreamingPlatform	5
SubscriptionPlan	12
Country	6
Subscriber	11
Review	36
Total	133

Note: The remaining individuals belong to the base TUV ontology (films, actors, directors, etc.) imported from the original Film Ontology.

Spreadsheet 1 — Platforms and Plans (platforms_and_plans.xlsx)

This spreadsheet defines streaming platforms, their available countries, offered plans, and plan attributes. Three Cellfie rules were used:

- **Rule 1 — StreamingPlatform:** Creates platform individuals with their URL and country/plan links.

```
Individual: @A*   Types: StreamingPlatform
Annotations: rdfs:label @B*
Facts:   platformURL @C*(xsd:string),
         availableInCountry @D*(mm:entityIRI),
         offersPlan @F*(mm:entityIRI)
```

- **Rule 2 — SubscriptionPlan:** Creates plan individuals with cost, stream limit, and ads flag.

```
Individual: @F*   Types: @G*
Facts:   subscriptionCost @H*(xsd:decimal),
         maxSimultaneousStreams @I*(xsd:integer),
         hasAds @J*(xsd:boolean)
```

- **Rule 3 — Country:** Creates country individuals with labels.

```
Individual: @D*   Types: Country
Annotations: rdfs:label @E*
```

Example assertions generated:

- platform_netflix availableInCountry country_romania
- platform_netflix offersPlan plan_net_prem
- plan_net_prem subscriptionCost 15.99
- plan_net_prem hasAds false

Spreadsheet 2 — TV Series Expansion (tv_series_expansion.xlsx)

This spreadsheet defines films, series, seasons, and episodes along with their genres, origin countries, and episode ordering. Six Cellfie rules were used:

- **Rule 1 — Artwork type and label:** Creates individuals with their class type and human-readable label.

```
Individual: @A*   Types: @C*
Annotations: rdfs:label @B*
```

- **Rule 2 — hasOriginCountry:** Links artworks to their country of origin.

```
Individual: @A*
Facts:   hasOriginCountry @D*(mm:SkipIfEmptyLocation)
```

- **Rule 3 — hasGenre:** Links artworks to their genre.

```
Individual: @A*
Facts:   hasGenre @E*(mm:SkipIfEmptyLocation)
```

- **Rule 4 — hasSeason:** Links series to their seasons.

```
Individual: @F*(mm:SkipIfEmptyLocation)
Facts:   hasSeason @A*
```

- **Rule 5 — hasEpisode:** Links seasons to their episodes.

```
Individual: @G*(mm:SkipIfEmptyLocation)
Facts:   hasEpisode @A*
```

- **Rule 6 — nextEpisode:** Chains episodes in order.

```
Individual:  @A*
Facts:  nextEpisode @H*(mm:SkipIfEmptyLocation)
```

Example assertions generated:

- series_chernobyl hasSeason season_chernobyl_1
- season_chernobyl_1 hasEpisode ep_cher_1
- ep_cher_1 nextEpisode ep_cher_2
- movie_inception hasGenre genre_science_fiction

Spreadsheet 3 — Users and Social (users_and_social.xlsx)

This spreadsheet defines subscribers, their subscriptions, social connections, viewing history, and reviews. Six Cellfie rules were used:

- **Rule 1 — Subscriber:** Creates subscriber individuals with age, country, platform and plan.

```
Individual:  @A*   Types:  Subscriber
Annotations:  rdfs:label @B*
Facts:  subscriberAge @C*(xsd:integer),
        locatedIn @D*(mm:SkipIfEmptyLocation),
        subscribedTo @E*(mm:SkipIfEmptyLocation),
        hasPlan @F*(mm:SkipIfEmptyLocation)
```

- **Rule 2 — friendOf:** Links subscribers to their friends.

```
Individual:  @A*
Facts:  friendOf @G*(mm:SkipIfEmptyLocation)
```

- **Rule 3 — watched:** Records which artworks a subscriber has watched.

```
Individual:  @A*
Facts:  watched @H*(mm:SkipIfEmptyLocation)
```

- **Rule 4 — likes:** Records which artworks a subscriber likes.

```
Individual:  @A*
Facts:  likes @I*(mm:SkipIfEmptyLocation)
```

- **Rule 5 — wroteReview:** Links subscribers to their written reviews.

```
Individual:  @A*
Facts:  wroteReview @J*(mm:SkipIfEmptyLocation)
```

- **Rule 6 — Review:** Creates review individuals with score and artwork link.

```
Individual:  @J*(mm:SkipIfEmptyLocation)   Types:  Review
Facts:  reviewScore @K*(xsd:integer),
        reviewsArtwork @H*(mm:SkipIfEmptyLocation)
```


Example assertions generated:

- `user_alex subscribedTo platform_netflix`
- `user_alex friendOf user_matei`
- `user_alex watched movie_inception`
- `rev_alex_1 reviewScore 9`
- `rev_alex_1 reviewsArtwork movie_dune_2`

Chapter 8

Check for Anomalies

After populating the ontology with individuals and defining all role constraints and SWRL rules, the ontology was validated using the **Pellet** reasoner in Protégé.

The following checks were performed:

- **Consistency check:** The reasoner confirmed the ontology is *consistent* — no contradictory axioms were found. No individual was simultaneously asserted to belong to two disjoint classes.
- **Unsatisfiable classes:** No unsatisfiable classes were detected. All defined classes (**SciFiFilm**, **EnsembleFilm**, **AwardWinningArtwork**) and inferred classes (**AdultSubscriber**, **PositiveReview**, **AdFreePlan**, **FullAccessSubscriber**) were satisfiable and populated with at least one individual after reasoning.
- **Unintended inferences:** The **AdultSubscriber** equivalence anomaly was detected and corrected during development — an accidental **EquivalentTo: Subscriber** axiom caused all subscribers to be classified as **AdultSubscriber**. This was removed, and the classification is now driven exclusively by the SWRL rule.
- **Property characteristic validation:** The symmetric property **friendOf** was verified to correctly infer reverse-direction assertions. The transitive property **sequel** was verified to infer indirect lineage between films. The irreflexive property **recommendsTo** correctly prevents self-recommendation.

No anomalies remain in the final version of the ontology.

Chapter 9

Ontology Evaluation

The ontology was evaluated by translating the competency questions defined in Chapter 1 into SPARQL queries and measuring the quality of the results using standard Information Retrieval metrics. Five queries were selected covering a range of complexity levels, and one query was executed both with and without the reasoner active to demonstrate the impact of inference.

Evaluation Metrics

The following metrics were used to assess each query:

- **Precision** – measures how many of the returned results are actually correct:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} = \frac{\text{Correct results returned}}{\text{Total results returned}}$$

- **Recall** – measures how many of the expected correct results were actually returned:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} = \frac{\text{Correct results returned}}{\text{Total expected results}}$$

- **F1 Score** – the harmonic mean of Precision and Recall, providing a single combined measure:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Accuracy** – the proportion of competency questions answered correctly overall:

$$\text{Accuracy} = \frac{\text{Correctly answered CQs}}{\text{Total CQs}}$$

In this context, a result is considered a *True Positive* if it appears in both the query output and the manually verified ground truth. A *False Positive* is a result returned by the query that should not have been. A *False Negative* is an expected result that the query failed to return.

Queries and Results

Q1 – All Films in the Catalog

```
SELECT ?film WHERE { ?film a :Film . }
```

Returns all individuals asserted as **Film**, including the 13 films added via Cellfie and the 5 films inherited from the base TUW ontology. Ground truth was verified manually from both sources.

Q2 – Subscribers Who Watched at Least One Sci-Fi Film

```
SELECT DISTINCT ?subscriber WHERE {  
  ?subscriber a :Subscriber ; :watched ?film .  
  ?film :hasGenre :genre_science_fiction .  
}
```

Joins subscriber watching history with genre assertions. Ground truth was verified by manually cross-referencing the `users_and_social.xlsx` data with films that have `genre_science_fiction`.

Q3 – Reviews with Score ≥ 9

```
SELECT ?review ?score WHERE {  
  ?review a :Review ; :reviewScore ?score .  
  FILTER(?score >= 9)  
}
```

Applies a numeric filter on a data property. Ground truth was computed by counting all review individuals with a `reviewScore` of 9 or 10 in the ontology.

Q4 – SciFiFilm Instances (Reasoning Validation)

```
SELECT ?film WHERE { ?film a :SciFiFilm . }
```

This query was executed twice – once without the reasoner and once with Pellet active in Snap SPARQL. `SciFiFilm` is a defined class; its instances are never asserted explicitly but are derived by the reasoner from the `hasGenre` restriction.

Q5 – Friend Pairs Who Co-Watched an Artwork

```
SELECT DISTINCT ?subA ?subB ?artwork WHERE {  
  ?subA :friendOf ?subB ; :watched ?artwork .  
  ?subB :watched ?artwork .  
  FILTER(STR(?subA) < STR(?subB))  
}
```

This query was executed in Snap SPARQL to leverage the symmetric inference of `friendOf`. Running it in the standard SPARQL tab (without inference) returned only 16 results because reverse-direction friend assertions are not explicitly stored – they are derived by the reasoner from the symmetric property characteristic. The ground truth of 34 includes inferred season-watching derived from SWRL Rule 3, which propagates series-level watching to season-level watching.

Results

Query	Exp.	Ret.	Correct	FP	FN	Prec.	Recall	F1
Q1 – All Films	18	18	18	0	0	1.000	1.000	1.000
Q2 – Watched sci-fi	8	8	8	0	0	1.000	1.000	1.000
Q3 – Reviews ≥ 9	29	29	29	0	0	1.000	1.000	1.000
Q4a – SciFiFilm (no reasoner)	6	0	0	0	6	–	0.000	0.000
Q4b – SciFiFilm (with reasoner)	6	6	6	0	0	1.000	1.000	1.000
Q5 – Co-watched (Snap SPARQL)	34	34	34	0	0	1.000	1.000	1.000

Summary:

- **Average Precision:** 1.000 (excluding Q4a where Precision is undefined)
- **Average Recall:** 1.000 (excluding Q4a)
- **Average F1:** 1.000 (excluding Q4a)
- **Overall Accuracy:** $5/5 = 100\%$ (treating Q4 as one CQ, answered correctly when the reasoner is active)
- **Accuracy without reasoner:** $4/5 = 80\%$

Reasoning Validation

The comparison between Q4a and Q4b directly demonstrates the necessity of the OWL reasoner. Without inference, the query returns zero results because `SciFiFilm` instances are never explicitly asserted – the class is defined by a restriction and populated entirely through classification. With Pellet active, six films are correctly identified.

Similarly, Q5 demonstrates the value of property characteristic reasoning: the `friendOf` symmetric property doubles the available friendship graph without requiring redundant data assertions, but only surfaces when the reasoner is engaged through Snap SPARQL.

These results confirm that the ontology is both correct and complete with respect to its competency questions, and that automated reasoning provides a meaningful extension of the explicitly asserted knowledge.

Bibliography

Appendix A

Your code

Intelligent Systems Group



```
1 @prefix : <http://semantics.id/ns/example/film#> .
2 @prefix owl: <http://www.w3.org/2002/07/owl#> .
3 @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
4 @prefix xml: <http://www.w3.org/XML/1998/namespace> .
5 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
6 @prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
7 @base <http://semantics.id/ns/example/film#> .
8
9 <http://semantics.id/ns/example/film#> rdf:type owl:Ontology ;
10                                     owl:versionIRI <http://semantics.id/ns/example/film/1.0.0> ;
11                                     <http://purl.org/dc/terms/contributor> "Filip Kovacevic" ,
12                                     "Laura Waltersdorfer" ,
13                                     "Marta Sabou" ;
14                                     <http://purl.org/dc/terms/created> "2022-09-30" ;
15                                     <http://purl.org/dc/terms/creator> "Fajar J. Ekaputra" ;
16                                     <http://purl.org/dc/terms/description> "An ontology about film and
17                                     ↪ related concepts developed for ISS course at TU Wien. Extended to include Streaming Platforms,
18                                     ↪ Subscriptions, and Awards." ;
19                                     <http://purl.org/dc/terms/title> "The Extended Streaming Film Ontology"
20                                     ↪ .
21
22 #####
23 # Annotation properties
24 #####
25
26 ### http://purl.org/dc/terms/contributor
27 <http://purl.org/dc/terms/contributor> rdf:type owl:AnnotationProperty .
28
29 ### http://purl.org/dc/terms/created
30 <http://purl.org/dc/terms/created> rdf:type owl:AnnotationProperty .
31
32 ### http://purl.org/dc/terms/creator
33 <http://purl.org/dc/terms/creator> rdf:type owl:AnnotationProperty .
34
35 ### http://purl.org/dc/terms/description
36 <http://purl.org/dc/terms/description> rdf:type owl:AnnotationProperty .
37
38 ### http://purl.org/dc/terms/title
39 <http://purl.org/dc/terms/title> rdf:type owl:AnnotationProperty .
40
41
42
43 ### http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled
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44 <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> rdf:type owl:AnnotationProperty .
45
46
47 ### http://www.w3.org/2002/07/owl#minCardinality
48 owl:minCardinality rdf:type owl:AnnotationProperty .
49
50
51 #####
52 # Datatypes
53 #####
54
55 ### http://www.w3.org/2001/XMLSchema#date
56 xsd:date rdf:type rdfs:Datatype .
57
58
59 #####
60 # Object Properties
61 #####
62
63 ### http://semantics.id/ns/example/film#availableInCountry
64 :availableInCountry rdf:type owl:ObjectProperty ;
65     rdfs:domain :StreamingPlatform ;
66     rdfs:range :Country ;
67     rdfs:label "availableInCountry" .
68
69
70 ### http://semantics.id/ns/example/film#availableOn
71 :availableOn rdf:type owl:ObjectProperty ;
72     rdfs:domain :Artwork ;
73     rdfs:range :StreamingPlatform ;
74     rdfs:label "availableOn" .
75
76
77 ### http://semantics.id/ns/example/film#belongsToSeries
78 :belongsToSeries rdf:type owl:ObjectProperty ;
79     rdfs:domain :Episode ;
80     rdfs:range :Series ;
81     rdfs:label "belongsToSeries" .
82
83
84 ### http://semantics.id/ns/example/film#coStarredWith
85 :coStarredWith rdf:type owl:ObjectProperty ,
86     owl:SymmetricProperty ;
87     rdfs:domain :Actor ;
88     rdfs:range :Actor ;
89     rdfs:label "coStarredWith" .
90
91
92 ### http://semantics.id/ns/example/film#coWatched
93 :coWatched rdf:type owl:ObjectProperty ,
94     owl:SymmetricProperty ;
95     rdfs:domain :Subscriber ;
96     rdfs:range :Subscriber ;
97     rdfs:label "coWatched" .
98
99
100 ### http://semantics.id/ns/example/film#friendOf
101 :friendOf rdf:type owl:ObjectProperty ;
102     rdfs:subPropertyOf <http://xmlns.com/foaf/0.1/knows> ;
103     rdf:type owl:SymmetricProperty ;
104     rdfs:domain <http://xmlns.com/foaf/0.1/Person> ;
105     rdfs:range <http://xmlns.com/foaf/0.1/Person> ;
106     rdfs:label "friendOf" .
107
108
109 ### http://semantics.id/ns/example/film#hasActor
110 :hasActor rdf:type owl:ObjectProperty ;
111     rdfs:subPropertyOf :hasPerformer ;
112     rdfs:domain :Film ;
113     rdfs:range :Actor ;
114     rdfs:label "hasActor" ;
115     owl:minCardinality 1 .
116
117
118 ### http://semantics.id/ns/example/film#hasAgeRating
119 :hasAgeRating rdf:type owl:ObjectProperty ,

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120         owl:FunctionalProperty ;
121         rdfs:domain :Artwork ;
122         rdfs:range :AgeRating ;
123         rdfs:label "hasAgeRating" .
124
125
126     ### http://semantics.id/ns/example/film#hasAward
127 :hasAward rdf:type owl:ObjectProperty ;
128         rdfs:domain :Artwork ;
129         rdfs:range :Award ;
130         rdfs:label "hasAward" .
131
132
133     ### http://semantics.id/ns/example/film#hasCinematographer
134 :hasCinematographer rdf:type owl:ObjectProperty ;
135         rdfs:subPropertyOf :hasCrew ;
136         rdfs:domain :Film ;
137         rdfs:range :Cinematographer ;
138         rdfs:label "hasCinematographer" .
139
140
141     ### http://semantics.id/ns/example/film#hasComposer
142 :hasComposer rdf:type owl:ObjectProperty ;
143         rdfs:subPropertyOf :hasCrew ;
144         rdfs:domain :Film ;
145         rdfs:range :Composer ;
146         rdfs:label "hasComposer" .
147
148
149     ### http://semantics.id/ns/example/film#hasCostumeDesigner
150 :hasCostumeDesigner rdf:type owl:ObjectProperty ;
151         rdfs:subPropertyOf :hasCrew ;
152         rdfs:domain :Film ;
153         rdfs:range :CostumeDesigner ;
154         rdfs:label "hasCostumeDesigner" .
155
156
157     ### http://semantics.id/ns/example/film#hasCrew
158 :hasCrew rdf:type owl:ObjectProperty ;
159         rdfs:domain :Film ;
160         rdfs:range <http://xmlns.com/foaf/0.1/Person> ;
161         rdfs:label "hasCrew" .
162
163
164     ### http://semantics.id/ns/example/film#hasDirector
165 :hasDirector rdf:type owl:ObjectProperty ;
166         rdfs:subPropertyOf :hasCrew ;
167         rdfs:domain :Film ;
168         rdfs:range :Director ;
169         rdfs:label "hasDirector" ;
170         owl:minCardinality 1 .
171
172
173     ### http://semantics.id/ns/example/film#hasEditor
174 :hasEditor rdf:type owl:ObjectProperty ;
175         rdfs:subPropertyOf :hasCrew ;
176         rdfs:domain :Film ;
177         rdfs:range :Editor ;
178         rdfs:label "hasEditor" .
179
180
181     ### http://semantics.id/ns/example/film#hasEpisode
182 :hasEpisode rdf:type owl:ObjectProperty ;
183         rdfs:domain :Season ;
184         rdfs:range :Episode ;
185         rdfs:label "hasEpisode" .
186
187
188     ### http://semantics.id/ns/example/film#hasFilmStudio
189 :hasFilmStudio rdf:type owl:ObjectProperty ;
190         rdfs:domain :Film ;
191         rdfs:range :FilmStudio ;
192         rdfs:label "hasFilmStudio" ;
193         owl:minCardinality 1 .
194
195

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196 ### http://semantics.id/ns/example/film#hasGenre
197 :hasGenre rdf:type owl:ObjectProperty ;
198         rdfs:domain :Film ;
199         rdfs:range :Genre ;
200         rdfs:label "hasGenre" ;
201         owl:minCardinality 1 .
202
203
204 ### http://semantics.id/ns/example/film#hasOriginCountry
205 :hasOriginCountry rdf:type owl:ObjectProperty ;
206         rdfs:domain :Artwork ;
207         rdfs:range :Country ;
208         rdfs:label "hasOriginCountry" .
209
210
211 ### http://semantics.id/ns/example/film#hasPerformer
212 :hasPerformer rdf:type owl:ObjectProperty ;
213         rdfs:domain :Artwork ;
214         rdfs:range :Performer ;
215         rdfs:label "hasPerformer" .
216
217
218 ### http://semantics.id/ns/example/film#hasPlan
219 :hasPlan rdf:type owl:ObjectProperty ;
220         rdfs:domain :Subscriber ;
221         rdfs:range :SubscriptionPlan ;
222         rdfs:label "hasPlan" .
223
224
225 ### http://semantics.id/ns/example/film#hasProducer
226 :hasProducer rdf:type owl:ObjectProperty ;
227         rdfs:subPropertyOf :hasCrew ;
228         rdfs:domain :Film ;
229         rdfs:range :Producer ;
230         rdfs:label "hasProducer" .
231
232
233 ### http://semantics.id/ns/example/film#hasScriptWriter
234 :hasScriptWriter rdf:type owl:ObjectProperty ;
235         rdfs:subPropertyOf :hasCrew ;
236         rdfs:domain :Film ;
237         rdfs:range :ScriptWriter ;
238         rdfs:label "hasScriptWriter" ;
239         owl:minCardinality 1 .
240
241
242 ### http://semantics.id/ns/example/film#hasSeason
243 :hasSeason rdf:type owl:ObjectProperty ;
244         rdfs:domain :Series ;
245         rdfs:range :Season ;
246         rdfs:label "hasSeason" .
247
248
249 ### http://semantics.id/ns/example/film#hasSoundtrack
250 :hasSoundtrack rdf:type owl:ObjectProperty ;
251         rdfs:domain :Film ;
252         rdfs:range :Soundtrack ;
253         rdfs:label "hasSoundtrack" .
254
255
256 ### http://semantics.id/ns/example/film#inProgress
257 :inProgress rdf:type owl:ObjectProperty ;
258         rdfs:domain :Subscriber ;
259         rdfs:range :Episode ;
260         rdfs:label "inProgress" .
261
262
263 ### http://semantics.id/ns/example/film#inSameUniverse
264 :inSameUniverse rdf:type owl:ObjectProperty ,
265                 owl:ReflexiveProperty ;
266         rdfs:domain :Artwork ;
267         rdfs:range :Artwork ;
268         rdfs:label "inSameUniverse" .
269
270
271 ### http://semantics.id/ns/example/film#likes

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272 :likes rdf:type owl:ObjectProperty ;
273     rdfs:domain :Subscriber ;
274     rdfs:range :Artwork ;
275     rdfs:label "likes" .
276
277
278 ### http://semantics.id/ns/example/film#locatedIn
279 :locatedIn rdf:type owl:ObjectProperty ,
280             owl:FunctionalProperty ;
281     rdfs:domain :Subscriber ;
282     rdfs:range :Country ;
283     rdfs:label "locatedIn" .
284
285
286 ### http://semantics.id/ns/example/film#nextEpisode
287 :nextEpisode rdf:type owl:ObjectProperty ,
288               owl:AsymmetricProperty ;
289     rdfs:domain :Episode ;
290     rdfs:range :Episode ;
291     rdfs:label "nextEpisode" .
292
293
294 ### http://semantics.id/ns/example/film#offersPlan
295 :offersPlan rdf:type owl:ObjectProperty ;
296     rdfs:domain :StreamingPlatform ;
297     rdfs:range :SubscriptionPlan ;
298     rdfs:label "offersPlan" .
299
300
301 ### http://semantics.id/ns/example/film#prequel
302 :prequel rdf:type owl:ObjectProperty ;
303     owl:inverseOf :sequel ;
304     rdf:type owl:TransitiveProperty ;
305     rdfs:domain :Film ;
306     rdfs:range :Film ;
307     rdfs:label "prequel" .
308
309
310 ### http://semantics.id/ns/example/film#recommendsTo
311 :recommendsTo rdf:type owl:ObjectProperty ,
312               owl:IrreflexiveProperty ;
313     rdfs:domain :Subscriber ;
314     rdfs:range :Subscriber ;
315     rdfs:label "recommendsTo" .
316
317
318 ### http://semantics.id/ns/example/film#reviewsArtwork
319 :reviewsArtwork rdf:type owl:ObjectProperty ;
320     rdfs:domain :Review ;
321     rdfs:range :Artwork ;
322     rdfs:label "reviewsArtwork" .
323
324
325 ### http://semantics.id/ns/example/film#sequel
326 :sequel rdf:type owl:ObjectProperty ,
327           owl:TransitiveProperty ;
328     rdfs:domain :Film ;
329     rdfs:range :Film ;
330     rdfs:label "sequel" .
331
332
333 ### http://semantics.id/ns/example/film#subscribedTo
334 :subscribedTo rdf:type owl:ObjectProperty ;
335     rdfs:domain :Subscriber ;
336     rdfs:range :StreamingPlatform ;
337     rdfs:label "subscribedTo" .
338
339
340 ### http://semantics.id/ns/example/film#watched
341 :watched rdf:type owl:ObjectProperty ;
342     rdfs:domain :Subscriber ;
343     rdfs:range :Artwork ;
344     rdfs:label "watched" .
345
346
347 ### http://semantics.id/ns/example/film#wroteReview

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348 :wroteReview rdf:type owl:ObjectProperty ,
349         owl:InverseFunctionalProperty ;
350         rdfs:domain :Subscriber ;
351         rdfs:range :Review ;
352         rdfs:label "wroteReview" .
353
354
355 ### http://xmlns.com/foaf/0.1/knows
356 <http://xmlns.com/foaf/0.1/knows> rdf:type owl:ObjectProperty ;
357         rdfs:domain <http://xmlns.com/foaf/0.1/Person> ;
358         rdfs:range <http://xmlns.com/foaf/0.1/Person> .
359
360
361 #####
362 # Data properties
363 #####
364
365 ### http://semantics.id/ns/example/film#awardYear
366 :awardYear rdf:type owl:DatatypeProperty ;
367         rdfs:domain :Award ;
368         rdfs:range xsd:integer ;
369         rdfs:label "awardYear" .
370
371
372 ### http://semantics.id/ns/example/film#budget
373 :budget rdf:type owl:DatatypeProperty ;
374         rdfs:domain :Film ;
375         rdfs:range xsd:float ;
376         rdfs:label "budget" .
377
378
379 ### http://semantics.id/ns/example/film#dateOfBirth
380 :dateOfBirth rdf:type owl:DatatypeProperty ,
381         owl:FunctionalProperty ;
382         rdfs:domain <http://xmlns.com/foaf/0.1/Person> ;
383         rdfs:range xsd:date ;
384         rdfs:label "dateOfBirth" .
385
386
387 ### http://semantics.id/ns/example/film#duration
388 :duration rdf:type owl:DatatypeProperty ;
389         rdfs:domain :Artwork ;
390         rdfs:range xsd:float ;
391         rdfs:label "duration" .
392
393
394 ### http://semantics.id/ns/example/film#episodeNumber
395 :episodeNumber rdf:type owl:DatatypeProperty ;
396         rdfs:domain :Episode ;
397         rdfs:range xsd:integer ;
398         rdfs:label "episodeNumber" .
399
400
401 ### http://semantics.id/ns/example/film#establishedDate
402 :establishedDate rdf:type owl:DatatypeProperty ,
403         owl:FunctionalProperty ;
404         rdfs:domain :FilmStudio ;
405         rdfs:range xsd:date ;
406         rdfs:label "establishedDate" .
407
408
409 ### http://semantics.id/ns/example/film#fullName
410 :fullName rdf:type owl:DatatypeProperty ;
411         rdfs:domain <http://xmlns.com/foaf/0.1/Person> ;
412         rdfs:range xsd:string ;
413         rdfs:label "fullName" .
414
415
416 ### http://semantics.id/ns/example/film#gender
417 :gender rdf:type owl:DatatypeProperty ,
418         owl:FunctionalProperty ;
419         rdfs:domain <http://xmlns.com/foaf/0.1/Person> ;
420         rdfs:range xsd:string ;
421         rdfs:label "gender" .
422
423

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424 ### http://semantics.id/ns/example/film#hasAds
425 :hasAds rdf:type owl:DatatypeProperty ;
426       rdfs:domain :SubscriptionPlan ;
427       rdfs:range xsd:boolean ;
428       rdfs:label "hasAds" .
429
430
431 ### http://semantics.id/ns/example/film#imdbRating
432 :imdbRating rdf:type owl:DatatypeProperty ;
433       rdfs:domain :Artwork ;
434       rdfs:range xsd:decimal ;
435       rdfs:label "imdbRating" .
436
437
438 ### http://semantics.id/ns/example/film#maxSimultaneousStreams
439 :maxSimultaneousStreams rdf:type owl:DatatypeProperty ;
440       rdfs:domain :SubscriptionPlan ;
441       rdfs:range xsd:integer ;
442       rdfs:label "maxSimultaneousStreams" .
443
444
445 ### http://semantics.id/ns/example/film#originalLanguage
446 :originalLanguage rdf:type owl:DatatypeProperty ;
447       rdfs:domain :Artwork ;
448       rdfs:range xsd:string ;
449       rdfs:label "originalLanguage" .
450
451
452 ### http://semantics.id/ns/example/film#platformURL
453 :platformURL rdf:type owl:DatatypeProperty ;
454       rdfs:domain :StreamingPlatform ;
455       rdfs:range xsd:string ;
456       rdfs:label "platformURL" .
457
458
459 ### http://semantics.id/ns/example/film#registrationDate
460 :registrationDate rdf:type owl:DatatypeProperty ;
461       rdfs:domain :Subscriber ;
462       rdfs:range xsd:date ;
463       rdfs:label "registrationDate" .
464
465
466 ### http://semantics.id/ns/example/film#releaseYear
467 :releaseYear rdf:type owl:DatatypeProperty ;
468       rdfs:domain :Film ;
469       rdfs:range xsd:integer ;
470       rdfs:label "releaseYear" .
471
472
473 ### http://semantics.id/ns/example/film#reviewScore
474 :reviewScore rdf:type owl:DatatypeProperty ;
475       rdfs:domain :Review ;
476       rdfs:range xsd:integer ;
477       rdfs:label "reviewScore" .
478
479
480 ### http://semantics.id/ns/example/film#seasonNumber
481 :seasonNumber rdf:type owl:DatatypeProperty ;
482       rdfs:domain :Season ;
483       rdfs:range xsd:integer ;
484       rdfs:label "seasonNumber" .
485
486
487 ### http://semantics.id/ns/example/film#subscriberAge
488 :subscriberAge rdf:type owl:DatatypeProperty ;
489       rdfs:domain :Subscriber ;
490       rdfs:range xsd:integer ;
491       rdfs:label "subscriberAge" .
492
493
494 ### http://semantics.id/ns/example/film#subscriptionCost
495 :subscriptionCost rdf:type owl:DatatypeProperty ;
496       rdfs:domain :SubscriptionPlan ;
497       rdfs:range xsd:decimal ;
498       rdfs:label "subscriptionCost" .
499

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500
501 ### http://semantics.id/ns/example/film#synopsis
502 :synopsis rdf:type owl:DatatypeProperty ;
503         rdfs:domain :Artwork ;
504         rdfs:range xsd:string ;
505         rdfs:label "synopsis" .
506
507
508 #####
509 # Classes
510 #####
511
512 ### http://semantics.id/ns/example/film#Actor
513 :Actor rdf:type owl:Class ;
514         rdfs:subClassOf :Performer ,
515         <http://xmlns.com/foaf/0.1/Person> ;
516         rdfs:label "Actor" .
517
518
519 ### http://semantics.id/ns/example/film#AdFreePlan
520 :AdFreePlan rdf:type owl:Class ;
521         rdfs:subClassOf :SubscriptionPlan .
522
523
524 ### http://semantics.id/ns/example/film#AdSupportedPlan
525 :AdSupportedPlan rdf:type owl:Class ;
526         rdfs:subClassOf :SubscriptionPlan ;
527         rdfs:label "AdSupportedPlan" .
528
529
530 ### http://semantics.id/ns/example/film#AdultSubscriber
531 :AdultSubscriber rdf:type owl:Class ;
532         rdfs:subClassOf :Subscriber .
533
534
535 ### http://semantics.id/ns/example/film#AgeRating
536 :AgeRating rdf:type owl:Class ;
537         rdfs:label "AgeRating" .
538
539
540 ### http://semantics.id/ns/example/film#Artwork
541 :Artwork rdf:type owl:Class ;
542         rdfs:label "Artwork" .
543
544
545 ### http://semantics.id/ns/example/film#Award
546 :Award rdf:type owl:Class ;
547         rdfs:label "Award" .
548
549
550 ### http://semantics.id/ns/example/film#AwardWinningArtwork
551 :AwardWinningArtwork rdf:type owl:Class ;
552         owl:equivalentClass [ owl:intersectionOf ( :Artwork
553                                     [ rdf:type owl:Restriction ;
554                                     owl:onProperty :hasAward ;
555                                     owl:someValuesFrom :Award
556                                     ]
557                                     ) ;
558         rdf:type owl:Class
559         ] ;
560         rdfs:label "AwardWinningArtwork" .
561
562
563 ### http://semantics.id/ns/example/film#BasicPlan
564 :BasicPlan rdf:type owl:Class ;
565         rdfs:subClassOf :SubscriptionPlan ;
566         rdfs:label "BasicPlan" .
567
568
569 ### http://semantics.id/ns/example/film#BestComedy
570 :BestComedy rdf:type owl:Class ;
571         rdfs:subClassOf :Emmy ;
572         rdfs:label "BestComedy" .
573
574
575 ### http://semantics.id/ns/example/film#BestDrama

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576 :BestDrama rdf:type owl:Class ;
577       rdfs:subClassOf :Emmy ;
578       rdfs:label "BestDrama" .
579
580
581 ### http://semantics.id/ns/example/film#Cinematographer
582 :Cinematographer rdf:type owl:Class ;
583       rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
584       rdfs:label "Cinematographer" .
585
586
587 ### http://semantics.id/ns/example/film#Composer
588 :Composer rdf:type owl:Class ;
589       rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
590       rdfs:label "Composer" .
591
592
593 ### http://semantics.id/ns/example/film#CostumeDesigner
594 :CostumeDesigner rdf:type owl:Class ;
595       rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
596       rdfs:label "CostumeDesigner" .
597
598
599 ### http://semantics.id/ns/example/film#Country
600 :Country rdf:type owl:Class ;
601       rdfs:label "Country" .
602
603
604 ### http://semantics.id/ns/example/film#Director
605 :Director rdf:type owl:Class ;
606       rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
607       rdfs:label "Director" .
608
609
610 ### http://semantics.id/ns/example/film#Editor
611 :Editor rdf:type owl:Class ;
612       rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
613       rdfs:label "Editor" .
614
615
616 ### http://semantics.id/ns/example/film#Emmy
617 :Emmy rdf:type owl:Class ;
618       rdfs:subClassOf :Award ;
619       rdfs:label "Emmy" .
620
621
622 ### http://semantics.id/ns/example/film#EnsembleFilm
623 :EnsembleFilm rdf:type owl:Class ;
624       owl:equivalentClass [ owl:intersectionOf ( :Film
625                               [ rdf:type owl:Restriction ;
626                                 owl:onProperty :hasActor ;
627                                 owl:minQualifiedCardinality "3"^^xsd:
628                                     ↪ nonNegativeInteger ;
629                                     owl:onClass :Actor
630                               ]
631                             ) ;
632       rdfs:label "EnsembleFilm" .
633
634
635
636 ### http://semantics.id/ns/example/film#Episode
637 :Episode rdf:type owl:Class ;
638       rdfs:subClassOf :Artwork ;
639       rdfs:label "Episode" .
640
641
642 ### http://semantics.id/ns/example/film#FamilyPlan
643 :FamilyPlan rdf:type owl:Class ;
644       rdfs:subClassOf :SubscriptionPlan ;
645       rdfs:label "FamilyPlan" .
646
647
648 ### http://semantics.id/ns/example/film#Film
649 :Film rdf:type owl:Class ;
650       rdfs:subClassOf :Artwork ;

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651         rdfs:label "Film" .
652
653
654     ### http://semantics.id/ns/example/film#FilmStudio
655     :FilmStudio rdf:type owl:Class ;
656         rdfs:label "FilmStudio" .
657
658
659     ### http://semantics.id/ns/example/film#FullAccessSubscriber
660     :FullAccessSubscriber rdf:type owl:Class ;
661         rdfs:subClassOf :Subscriber .
662
663
664     ### http://semantics.id/ns/example/film#Genre
665     :Genre rdf:type owl:Class ;
666         rdfs:label "Genre" .
667
668
669     ### http://semantics.id/ns/example/film#GoldenGlobe
670     :GoldenGlobe rdf:type owl:Class ;
671         rdfs:subClassOf :Award ;
672         rdfs:label "GoldenGlobe" .
673
674
675     ### http://semantics.id/ns/example/film#Miniseries
676     :Miniseries rdf:type owl:Class ;
677         rdfs:subClassOf :Series ;
678         rdfs:label "Miniseries" .
679
680
681     ### http://semantics.id/ns/example/film#Oscar
682     :Oscar rdf:type owl:Class ;
683         rdfs:subClassOf :Award ;
684         rdfs:label "Oscar" .
685
686
687     ### http://semantics.id/ns/example/film#PalmeDOr
688     :PalmeDOr rdf:type owl:Class ;
689         rdfs:subClassOf :Award ;
690         rdfs:label "PalmeDOr" .
691
692
693     ### http://semantics.id/ns/example/film#Performer
694     :Performer rdf:type owl:Class ;
695         rdfs:label "Performer" .
696
697
698     ### http://semantics.id/ns/example/film#PositiveReview
699     :PositiveReview rdf:type owl:Class ;
700         rdfs:subClassOf :Review .
701
702
703     ### http://semantics.id/ns/example/film#PremiumPlan
704     :PremiumPlan rdf:type owl:Class ;
705         rdfs:subClassOf :SubscriptionPlan ;
706         rdfs:label "PremiumPlan" .
707
708
709     ### http://semantics.id/ns/example/film#Producer
710     :Producer rdf:type owl:Class ;
711         rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
712         rdfs:label "Producer" .
713
714
715     ### http://semantics.id/ns/example/film#Review
716     :Review rdf:type owl:Class ;
717         rdfs:label "Review" .
718
719
720     ### http://semantics.id/ns/example/film#SciFiFilm
721     :SciFiFilm rdf:type owl:Class ;
722         owl:equivalentClass [ owl:intersectionOf ( :Film
723             [ rdf:type owl:Restriction ;
724               owl:onProperty :hasGenre ;
725               owl:hasValue :genre_science_fiction
726             ]

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727         ) ;
728         rdf:type owl:Class
729     ] ;
730     rdfs:label "SciFiFilm" .
731
732
733     ### http://semantics.id/ns/example/film#ScriptWriter
734     :ScriptWriter rdf:type owl:Class ;
735         rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
736         rdfs:label "ScriptWriter" .
737
738
739     ### http://semantics.id/ns/example/film#Season
740     :Season rdf:type owl:Class ;
741         rdfs:subClassOf :Artwork ;
742         rdfs:label "Season" .
743
744
745     ### http://semantics.id/ns/example/film#Series
746     :Series rdf:type owl:Class ;
747         rdfs:subClassOf :Artwork ;
748         rdfs:label "Series" .
749
750
751     ### http://semantics.id/ns/example/film#ShortFilm
752     :ShortFilm rdf:type owl:Class ;
753         rdfs:subClassOf :Film ;
754         rdfs:label "ShortFilm" .
755
756
757     ### http://semantics.id/ns/example/film#Soundtrack
758     :Soundtrack rdf:type owl:Class ;
759         rdfs:subClassOf :Artwork ;
760         rdfs:label "Soundtrack" .
761
762
763     ### http://semantics.id/ns/example/film#StreamingPlatform
764     :StreamingPlatform rdf:type owl:Class ;
765         rdfs:label "StreamingPlatform" .
766
767
768     ### http://semantics.id/ns/example/film#StudentPlan
769     :StudentPlan rdf:type owl:Class ;
770         rdfs:subClassOf :SubscriptionPlan ;
771         rdfs:label "StudentPlan" .
772
773
774     ### http://semantics.id/ns/example/film#Subscriber
775     :Subscriber rdf:type owl:Class ;
776         rdfs:subClassOf <http://xmlns.com/foaf/0.1/Person> ;
777         rdfs:label "Subscriber" .
778
779
780     ### http://semantics.id/ns/example/film#SubscriptionPlan
781     :SubscriptionPlan rdf:type owl:Class ;
782         rdfs:label "SubscriptionPlan" .
783
784
785     ### http://semantics.id/ns/example/film#Trailer
786     :Trailer rdf:type owl:Class ;
787         rdfs:subClassOf :Artwork ;
788         rdfs:label "Trailer" .
789
790
791     ### http://semantics.id/ns/example/film#VoiceActor
792     :VoiceActor rdf:type owl:Class ;
793         rdfs:subClassOf :Actor ;
794         rdfs:label "VoiceActor" .
795
796
797     ### http://xmlns.com/foaf/0.1/Person
798     <http://xmlns.com/foaf/0.1/Person> rdf:type owl:Class .
799
800
801     #####
802     # Individuals

```

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803 #####
804
805 ### http://semantics.id/ns/example#EntertainmentOne
806 <http://semantics.id/ns/example#EntertainmentOne> rdf:type owl:NamedIndividual ,
807                                     :FilmStudio ;
808                                     :establishedDate "1973-01-01"^^xsd:date ;
809                                     rdfs:label "Entertainment One" .
810
811
812 ### http://semantics.id/ns/example#MandevilleFilms
813 <http://semantics.id/ns/example#MandevilleFilms> rdf:type owl:NamedIndividual ,
814                                     :FilmStudio ;
815                                     :establishedDate "1955-01-01"^^xsd:date ;
816                                     rdfs:label "Mandeville Films" .
817
818
819 ### http://semantics.id/ns/example#WaltDisneyPictures
820 <http://semantics.id/ns/example#WaltDisneyPictures> rdf:type owl:NamedIndividual ,
821                                     :FilmStudio ;
822                                     :establishedDate "1985-06-21"^^xsd:date ;
823                                     rdfs:label "Walt Disney Pictures" .
824
825
826 ### http://semantics.id/ns/example#alicia_vikander
827 <http://semantics.id/ns/example#alicia_vikander> rdf:type owl:NamedIndividual ,
828                                     :Actor ;
829                                     :fullName "Alicia Vikander" ;
830                                     :gender "female" .
831
832
833 ### http://semantics.id/ns/example#bill_condon
834 <http://semantics.id/ns/example#bill_condon> rdf:type owl:NamedIndividual ,
835                                     :Director ;
836                                     :fullName "Bill Condon" ;
837                                     :gender "male" .
838
839
840 ### http://semantics.id/ns/example#dan_stevens
841 <http://semantics.id/ns/example#dan_stevens> rdf:type owl:NamedIndividual ,
842                                     :Actor ;
843                                     :dateOfBirth "1982-10-10"^^xsd:date ;
844                                     :fullName "Dan Stevens" ;
845                                     :gender "male" .
846
847
848 ### http://semantics.id/ns/example#emma_watson
849 <http://semantics.id/ns/example#emma_watson> rdf:type owl:NamedIndividual ,
850                                     :Actor ;
851                                     :dateOfBirth "1990-04-15"^^xsd:date ;
852                                     :fullName "Emma Watson" ;
853                                     :gender "female" .
854
855
856 ### http://semantics.id/ns/example#ewan_mcgregor
857 <http://semantics.id/ns/example#ewan_mcgregor> rdf:type owl:NamedIndividual ,
858                                     :Actor ;
859                                     :dateOfBirth "1971-03-31"^^xsd:date ;
860                                     :fullName "Ewan McGregor" ;
861                                     :gender "male" .
862
863
864 ### http://semantics.id/ns/example#film_1
865 <http://semantics.id/ns/example#film_1> rdf:type owl:NamedIndividual ,
866                                     :Film ;
867                                     :hasActor <http://semantics.id/ns/example#ewan_mcgregor> ,
868                                     <http://semantics.id/ns/example#hayley_atwell> ;
869                                     :hasDirector <http://semantics.id/ns/example#marc_forster> ;
870                                     :hasFilmStudio <http://semantics.id/ns/example#WaltDisneyPictures> ;
871                                     :hasGenre :genre_animation ,
872                                     :genre_family ;
873                                     :hasScriptWriter <http://semantics.id/ns/example#writer_1> ,
874                                     <http://semantics.id/ns/example#writer_2> ;
875                                     :releaseYear 2018 ;
876                                     rdfs:label "Christopher Robin" .
877
878

```

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879 ### http://semantics.id/ns/example#film_2
880 <http://semantics.id/ns/example#film_2> rdf:type owl:NamedIndividual ,
881                                     :Film ;
882                                     :hasActor <http://semantics.id/ns/example#alicia_vikander> ,
883                                     <http://semantics.id/ns/example#ewan_mcgregor> ;
884                                     :hasDirector <http://semantics.id/ns/example#julius_avery> ;
885                                     :hasFilmStudio <http://semantics.id/ns/example#EntertainmentOne> ;
886                                     :hasGenre :genre_action ,
887                                     :genre_drama ;
888                                     :hasScriptWriter <http://semantics.id/ns/example#julius_avery> ,
889                                     <http://semantics.id/ns/example#writer_3> ;
890                                     :releaseYear 2014 ;
891                                     rdfs:label "Son of a Gun" .
892
893
894 ### http://semantics.id/ns/example#film_3
895 <http://semantics.id/ns/example#film_3> rdf:type owl:NamedIndividual ,
896                                     :Film ;
897                                     :hasActor <http://semantics.id/ns/example#dan_stevens> ,
898                                     <http://semantics.id/ns/example#emma_watson> ,
899                                     <http://semantics.id/ns/example#ewan_mcgregor> ;
900                                     :hasDirector <http://semantics.id/ns/example#bill_condon> ;
901                                     :hasFilmStudio <http://semantics.id/ns/example#WaltDisneyPictures> ;
902                                     :hasGenre :genre_family ,
903                                     :genre_romance ;
904                                     :hasScriptWriter <http://semantics.id/ns/example#writer_4> ,
905                                     <http://semantics.id/ns/example#writer_5> ;
906                                     :hasSoundtrack :Belle ;
907                                     :releaseYear 2017 ;
908                                     rdfs:label "Beauty and the Beast" .
909
910
911 ### http://semantics.id/ns/example#film_4
912 <http://semantics.id/ns/example#film_4> rdf:type owl:NamedIndividual ,
913                                     :Film ;
914                                     :hasActor :zendaya ;
915                                     :hasComposer :hans_zimmer ;
916                                     :hasGenre :genre_science_fiction ;
917                                     :releaseYear 2021 ;
918                                     rdfs:label "Dune" .
919
920
921 ### http://semantics.id/ns/example#film_5
922 <http://semantics.id/ns/example#film_5> rdf:type owl:NamedIndividual ,
923                                     :Film ;
924                                     :hasActor :kyle_macclachlan ;
925                                     :hasGenre :genre_science_fiction ;
926                                     :releaseYear 1984 ;
927                                     rdfs:label "Dune" .
928
929
930 ### http://semantics.id/ns/example#hayley_atwell
931 <http://semantics.id/ns/example#hayley_atwell> rdf:type owl:NamedIndividual ,
932                                     :Actor ;
933                                     :dateOfBirth "1982-04-05"^^xsd:date ;
934                                     :fullName "Hayley Atwell" ;
935                                     :gender "female" .
936
937
938 ### http://semantics.id/ns/example#julius_avery
939 <http://semantics.id/ns/example#julius_avery> rdf:type owl:NamedIndividual ,
940                                     :Director ,
941                                     :ScriptWriter ;
942                                     :fullName "Julius Avery" ;
943                                     :gender "male" .
944
945
946 ### http://semantics.id/ns/example#marc_forster
947 <http://semantics.id/ns/example#marc_forster> rdf:type owl:NamedIndividual ,
948                                     :Director ;
949                                     :fullName "Marc Forster" ;
950                                     :gender "male" .
951
952
953 ### http://semantics.id/ns/example#writer_1
954 <http://semantics.id/ns/example#writer_1> rdf:type owl:NamedIndividual ,

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955         :ScriptWriter ;
956         :dateOfBirth "1966-06-07"^^xsd:date ;
957         :fullName "Tom McCarthy" ;
958         :gender "male" .
959
960
961 ### http://semantics.id/ns/example#writer_2
962 <http://semantics.id/ns/example#writer_2> rdf:type owl:NamedIndividual ,
963         :ScriptWriter ;
964         :fullName "Alex Ross" .
965
966
967 ### http://semantics.id/ns/example#writer_3
968 <http://semantics.id/ns/example#writer_3> rdf:type owl:NamedIndividual ,
969         :ScriptWriter ;
970         :fullName "John Collee" ;
971         :gender "male" .
972
973
974 ### http://semantics.id/ns/example#writer_4
975 <http://semantics.id/ns/example#writer_4> rdf:type owl:NamedIndividual ,
976         :ScriptWriter ;
977         :fullName "Stephen Chbosky" ;
978         :gender "male" .
979
980
981 ### http://semantics.id/ns/example#writer_5
982 <http://semantics.id/ns/example#writer_5> rdf:type owl:NamedIndividual ,
983         :ScriptWriter ;
984         :fullName "Evan Spiliotopoulos" .
985
986
987 ### http://semantics.id/ns/example/film#Belle
988 :Belle rdf:type owl:NamedIndividual ,
989         :Soundtrack ;
990         :hasPerformer :alan_menken ;
991         rdfs:label "Belle (Soundtrack Version)" ;
992         rdfs:seeAlso "https://www.youtube.com/watch?v=piYIZaoBJMQ" .
993
994
995 ### http://semantics.id/ns/example/film#alan_menken
996 :alan_menken rdf:type owl:NamedIndividual ,
997         :Composer ,
998         :Performer ;
999         :fullName "Alan Menken" ;
1000         :gender "male" .
1001
1002
1003 ### http://semantics.id/ns/example/film#country_germany
1004 :country_germany rdf:type owl:NamedIndividual ,
1005         :Country ;
1006         rdfs:label "Germany" .
1007
1008
1009 ### http://semantics.id/ns/example/film#country_japan
1010 :country_japan rdf:type owl:NamedIndividual ,
1011         :Country ;
1012         rdfs:label "Japan" .
1013
1014
1015 ### http://semantics.id/ns/example/film#country_romania
1016 :country_romania rdf:type owl:NamedIndividual ,
1017         :Country ;
1018         rdfs:label "Romania" .
1019
1020
1021 ### http://semantics.id/ns/example/film#country_spain
1022 :country_spain rdf:type owl:NamedIndividual ,
1023         :Country ;
1024         rdfs:label "Spain" .
1025
1026
1027 ### http://semantics.id/ns/example/film#country_uk
1028 :country_uk rdf:type owl:NamedIndividual ,
1029         :Country ;
1030         rdfs:label "United Kingdom" .

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1031
1032
1033 ### http://semantics.id/ns/example/film#country_usa
1034 :country_usa rdf:type owl:NamedIndividual ,
1035               :Country ;
1036         rdfs:label "USA" .
1037
1038
1039 ### http://semantics.id/ns/example/film#ep_arc_1
1040 :ep_arc_1 rdf:type owl:NamedIndividual ,
1041             :Episode ;
1042         :nextEpisode :ep_arc_2 ;
1043         rdfs:label "Welcome to the Playground" .
1044
1045
1046 ### http://semantics.id/ns/example/film#ep_arc_2
1047 :ep_arc_2 rdf:type owl:NamedIndividual ,
1048             :Episode ;
1049         :nextEpisode :ep_arc_3 ;
1050         rdfs:label "Some Mysteries Are Better Left Unsolved" .
1051
1052
1053 ### http://semantics.id/ns/example/film#ep_arc_3
1054 :ep_arc_3 rdf:type owl:NamedIndividual ,
1055             :Episode ;
1056         :nextEpisode :ep_arc_4 ;
1057         rdfs:label "The Base Violence Necessary for Change" .
1058
1059
1060 ### http://semantics.id/ns/example/film#ep_arc_4
1061 :ep_arc_4 rdf:type owl:NamedIndividual ,
1062             :Episode ;
1063         :nextEpisode :ep_arc_5 ;
1064         rdfs:label "Happy Progress Day!" .
1065
1066
1067 ### http://semantics.id/ns/example/film#ep_arc_5
1068 :ep_arc_5 rdf:type owl:NamedIndividual ,
1069             :Episode ;
1070         :nextEpisode :ep_arc_6 ;
1071         rdfs:label "Everybody Wants to Be My Enemy" .
1072
1073
1074 ### http://semantics.id/ns/example/film#ep_arc_6
1075 :ep_arc_6 rdf:type owl:NamedIndividual ,
1076             :Episode ;
1077         :nextEpisode :ep_arc_7 ;
1078         rdfs:label "When These Walls Come Tumbling Down" .
1079
1080
1081 ### http://semantics.id/ns/example/film#ep_arc_7
1082 :ep_arc_7 rdf:type owl:NamedIndividual ,
1083             :Episode ;
1084         :nextEpisode :ep_arc_8 ;
1085         rdfs:label "The Boy Savior" .
1086
1087
1088 ### http://semantics.id/ns/example/film#ep_arc_8
1089 :ep_arc_8 rdf:type owl:NamedIndividual ,
1090             :Episode ;
1091         :nextEpisode :ep_arc_9 ;
1092         rdfs:label "Oil and Water" .
1093
1094
1095 ### http://semantics.id/ns/example/film#ep_arc_9
1096 :ep_arc_9 rdf:type owl:NamedIndividual ,
1097             :Episode ;
1098         rdfs:label "The Monster You Created" .
1099
1100
1101 ### http://semantics.id/ns/example/film#ep_bcs_101
1102 :ep_bcs_101 rdf:type owl:NamedIndividual ,
1103              :Episode ;
1104         :nextEpisode :ep_bcs_102 ;
1105         rdfs:label "Uno" .
1106

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1107
1108 ### http://semantics.id/ns/example/film#ep_bcs_102
1109 :ep_bcs_102 rdf:type owl:NamedIndividual ,
1110             :Episode ;
1111             :nextEpisode :ep_bcs_103 ;
1112             rdfs:label "Mijo" .
1113
1114
1115 ### http://semantics.id/ns/example/film#ep_bcs_103
1116 :ep_bcs_103 rdf:type owl:NamedIndividual ,
1117             :Episode ;
1118             :nextEpisode :ep_bcs_104 ;
1119             rdfs:label "Nacho" .
1120
1121
1122 ### http://semantics.id/ns/example/film#ep_bcs_104
1123 :ep_bcs_104 rdf:type owl:NamedIndividual ,
1124             :Episode ;
1125             :nextEpisode :ep_bcs_105 ;
1126             rdfs:label "Hero" .
1127
1128
1129 ### http://semantics.id/ns/example/film#ep_bcs_105
1130 :ep_bcs_105 rdf:type owl:NamedIndividual ,
1131             :Episode ;
1132             :nextEpisode :ep_bcs_106 ;
1133             rdfs:label "Alpine Shepherd Boy" .
1134
1135
1136 ### http://semantics.id/ns/example/film#ep_bcs_106
1137 :ep_bcs_106 rdf:type owl:NamedIndividual ,
1138             :Episode ;
1139             :nextEpisode :ep_bcs_107 ;
1140             rdfs:label "Five-0" .
1141
1142
1143 ### http://semantics.id/ns/example/film#ep_bcs_107
1144 :ep_bcs_107 rdf:type owl:NamedIndividual ,
1145             :Episode ;
1146             :nextEpisode :ep_bcs_108 ;
1147             rdfs:label "Bingo" .
1148
1149
1150 ### http://semantics.id/ns/example/film#ep_bcs_108
1151 :ep_bcs_108 rdf:type owl:NamedIndividual ,
1152             :Episode ;
1153             :nextEpisode :ep_bcs_109 ;
1154             rdfs:label "RICO" .
1155
1156
1157 ### http://semantics.id/ns/example/film#ep_bcs_109
1158 :ep_bcs_109 rdf:type owl:NamedIndividual ,
1159             :Episode ;
1160             :nextEpisode :ep_bcs_110 ;
1161             rdfs:label "Pimento" .
1162
1163
1164 ### http://semantics.id/ns/example/film#ep_bcs_110
1165 :ep_bcs_110 rdf:type owl:NamedIndividual ,
1166             :Episode ;
1167             rdfs:label "Marco" .
1168
1169
1170 ### http://semantics.id/ns/example/film#ep_cher_1
1171 :ep_cher_1 rdf:type owl:NamedIndividual ,
1172            :Episode ;
1173            :nextEpisode :ep_cher_2 ;
1174            rdfs:label "1:23:45" .
1175
1176
1177 ### http://semantics.id/ns/example/film#ep_cher_2
1178 :ep_cher_2 rdf:type owl:NamedIndividual ,
1179            :Episode ;
1180            :nextEpisode :ep_cher_3 ;
1181            rdfs:label "Please Remain Calm" .
1182

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1183
1184 ### http://semantics.id/ns/example/film#ep_cher_3
1185 :ep_cher_3 rdf:type owl:NamedIndividual ,
1186             :Episode ;
1187             :nextEpisode :ep_cher_4 ;
1188             rdfs:label "Open Wide O Earth" .
1189
1190
1191 ### http://semantics.id/ns/example/film#ep_cher_4
1192 :ep_cher_4 rdf:type owl:NamedIndividual ,
1193             :Episode ;
1194             :nextEpisode :ep_cher_5 ;
1195             rdfs:label "The Happiness of All Mankind" .
1196
1197
1198 ### http://semantics.id/ns/example/film#ep_cher_5
1199 :ep_cher_5 rdf:type owl:NamedIndividual ,
1200             :Episode ;
1201             rdfs:label "Vichnaya Pamyat" .
1202
1203
1204 ### http://semantics.id/ns/example/film#ep_mh_1
1205 :ep_mh_1 rdf:type owl:NamedIndividual ,
1206           :Episode ;
1207           :nextEpisode :ep_mh_2 ;
1208           rdfs:label "Efektuar lo acordado" .
1209
1210
1211 ### http://semantics.id/ns/example/film#ep_mh_2
1212 :ep_mh_2 rdf:type owl:NamedIndividual ,
1213           :Episode ;
1214           :nextEpisode :ep_mh_3 ;
1215           rdfs:label "Imprudencias letales" .
1216
1217
1218 ### http://semantics.id/ns/example/film#ep_mh_3
1219 :ep_mh_3 rdf:type owl:NamedIndividual ,
1220           :Episode ;
1221           :nextEpisode :ep_mh_4 ;
1222           rdfs:label "Error de disparo" .
1223
1224
1225 ### http://semantics.id/ns/example/film#ep_mh_4
1226 :ep_mh_4 rdf:type owl:NamedIndividual ,
1227           :Episode ;
1228           :nextEpisode :ep_mh_5 ;
1229           rdfs:label "Caballo de Troya" .
1230
1231
1232 ### http://semantics.id/ns/example/film#ep_mh_5
1233 :ep_mh_5 rdf:type owl:NamedIndividual ,
1234           :Episode ;
1235           rdfs:label "El dia de la marmota" .
1236
1237
1238 ### http://semantics.id/ns/example/film#genre_action
1239 :genre_action rdf:type owl:NamedIndividual ,
1240               :Genre ;
1241               rdfs:label "Action" .
1242
1243
1244 ### http://semantics.id/ns/example/film#genre_animation
1245 :genre_animation rdf:type owl:NamedIndividual ,
1246                   :Genre ;
1247                   rdfs:label "Animation" .
1248
1249
1250 ### http://semantics.id/ns/example/film#genre_drama
1251 :genre_drama rdf:type owl:NamedIndividual ,
1252              :Genre ;
1253              rdfs:label "Drama" .
1254
1255
1256 ### http://semantics.id/ns/example/film#genre_family
1257 :genre_family rdf:type owl:NamedIndividual ,
1258               :Genre ;

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1259         rdfs:label "Family" .
1260
1261
1262     ### http://semantics.id/ns/example/film#genre_romance
1263     :genre_romance rdf:type owl:NamedIndividual ,
1264                     :Genre ;
1265     rdfs:label "Romance" .
1266
1267
1268     ### http://semantics.id/ns/example/film#genre_science_fiction
1269     :genre_science_fiction rdf:type owl:NamedIndividual ,
1270                                   :Genre ;
1271     rdfs:label "Science Fiction"@en .
1272
1273
1274     ### http://semantics.id/ns/example/film#hans_zimmer
1275     :hans_zimmer rdf:type owl:NamedIndividual ,
1276                       :Composer ;
1277     :fullName "Hans Zimmer" .
1278
1279
1280     ### http://semantics.id/ns/example/film#kyle_maciachlan
1281     :kyle_maciachlan rdf:type owl:NamedIndividual ,
1282                          :Actor ;
1283     :dateOfBirth "1959-02-22"^^xsd:date ;
1284     :fullName "Kyle Merritt MacLachlan" ;
1285     :gender "male" .
1286
1287
1288     ### http://semantics.id/ns/example/film#movie_432
1289     :movie_432 rdf:type owl:NamedIndividual ,
1290                   :Film ;
1291     :hasGenre :genre_drama ;
1292     :hasOriginCountry :country_romania ;
1293     rdfs:label "4 Months 3 Weeks and 2 Days" .
1294
1295
1296     ### http://semantics.id/ns/example/film#movie_barbie
1297     :movie_barbie rdf:type owl:NamedIndividual ,
1298                          :Film ;
1299     :hasGenre :genre_family ;
1300     :hasOriginCountry :country_usa ;
1301     rdfs:label "Barbie" .
1302
1303
1304     ### http://semantics.id/ns/example/film#movie_chuck
1305     :movie_chuck rdf:type owl:NamedIndividual ,
1306                       :Film ;
1307     :hasGenre :genre_action ;
1308     :hasOriginCountry :country_romania ;
1309     rdfs:label "Chuck Norris vs Communism" .
1310
1311
1312     ### http://semantics.id/ns/example/film#movie_colectiv
1313     :movie_colectiv rdf:type owl:NamedIndividual ,
1314                          :Film ;
1315     :hasGenre :genre_drama ;
1316     :hasOriginCountry :country_romania ;
1317     rdfs:label "Collective" .
1318
1319
1320     ### http://semantics.id/ns/example/film#movie_dark_knight
1321     :movie_dark_knight rdf:type owl:NamedIndividual ,
1322                              :Film ;
1323     :hasGenre :genre_action ;
1324     :hasOriginCountry :country_usa ;
1325     rdfs:label "The Dark Knight" .
1326
1327
1328     ### http://semantics.id/ns/example/film#movie_dune_2
1329     :movie_dune_2 rdf:type owl:NamedIndividual ,
1330                     :Film ;
1331     :hasGenre :genre_science_fiction ;
1332     :hasOriginCountry :country_usa ;
1333     rdfs:label "Dune: Part Two" .
1334

```



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1335
1336 ### http://semantics.id/ns/example/film#movie_gadiator
1337 :movie_gadiator rdf:type owl:NamedIndividual ,
1338                  :Film ;
1339                  :hasGenre :genre_action ;
1340                  :hasOriginCountry :country_uk ;
1341                  rdfs:label "Gladiator" .
1342
1343
1344 ### http://semantics.id/ns/example/film#movie_inception
1345 :movie_inception rdf:type owl:NamedIndividual ,
1346                  :Film ;
1347                  :hasGenre :genre_science_fiction ;
1348                  :hasOriginCountry :country_usa ;
1349                  rdfs:label "Inception" .
1350
1351
1352 ### http://semantics.id/ns/example/film#movie_interstellar
1353 :movie_interstellar rdf:type owl:NamedIndividual ,
1354                       :Film ;
1355                       :hasGenre :genre_science_fiction ;
1356                       :hasOriginCountry :country_usa ;
1357                       rdfs:label "Interstellar" .
1358
1359
1360 ### http://semantics.id/ns/example/film#movie_matrix
1361 :movie_matrix rdf:type owl:NamedIndividual ,
1362                  :Film ;
1363                  :hasGenre :genre_science_fiction ;
1364                  :hasOriginCountry :country_usa ;
1365                  rdfs:label "The Matrix" .
1366
1367
1368 ### http://semantics.id/ns/example/film#movie_oppenheimer
1369 :movie_oppenheimer rdf:type owl:NamedIndividual ,
1370                       :Film ;
1371                       :hasGenre :genre_drama ;
1372                       :hasOriginCountry :country_usa ;
1373                       rdfs:label "Oppenheimer" .
1374
1375
1376 ### http://semantics.id/ns/example/film#movie_parasite
1377 :movie_parasite rdf:type owl:NamedIndividual ,
1378                  :Film ;
1379                  :hasGenre :genre_drama ;
1380                  :hasOriginCountry :country_japan ;
1381                  rdfs:label "Parasite" .
1382
1383
1384 ### http://semantics.id/ns/example/film#movie_spirited
1385 :movie_spirited rdf:type owl:NamedIndividual ,
1386                  :Film ;
1387                  :hasGenre :genre_animation ;
1388                  :hasOriginCountry :country_japan ;
1389                  rdfs:label "Spirited Away" .
1390
1391
1392 ### http://semantics.id/ns/example/film#plan_dis_basic
1393 :plan_dis_basic rdf:type owl:NamedIndividual ,
1394                  :BasicPlan ;
1395                  :hasAds "false"^^xsd:boolean ;
1396                  :maxSimultaneousStreams 2 ;
1397                  :subscriptionCost 7.99 .
1398
1399
1400 ### http://semantics.id/ns/example/film#plan_dis_fam
1401 :plan_dis_fam rdf:type owl:NamedIndividual ,
1402                 :FamilyPlan ;
1403                 :hasAds "false"^^xsd:boolean ;
1404                 :maxSimultaneousStreams 4 ;
1405                 :subscriptionCost 13.99 .
1406
1407
1408 ### http://semantics.id/ns/example/film#plan_hulu_ad
1409 :plan_hulu_ad rdf:type owl:NamedIndividual ,
1410                 :AdSupportedPlan ;

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1411         :hasAds "true"^^xsd:boolean ;
1412         :maxSimultaneousStreams 2 ;
1413         :subscriptionCost 7.99 .
1414
1415
1416 ### http://semantics.id/ns/example/film#plan_hulu_basic
1417 :plan_hulu_basic rdf:type owl:NamedIndividual ,
1418                 :BasicPlan ;
1419         :hasAds "false"^^xsd:boolean ;
1420         :maxSimultaneousStreams 2 ;
1421         :subscriptionCost 12.99 .
1422
1423
1424 ### http://semantics.id/ns/example/film#plan_hulu_stu
1425 :plan_hulu_stu rdf:type owl:NamedIndividual ,
1426                 :StudentPlan ;
1427         :hasAds "true"^^xsd:boolean ;
1428         :maxSimultaneousStreams 1 ;
1429         :subscriptionCost 1.99 ,
1430                 4.99 .
1431
1432
1433 ### http://semantics.id/ns/example/film#plan_max_ad
1434 :plan_max_ad rdf:type owl:NamedIndividual ,
1435                 :AdSupportedPlan ;
1436         :hasAds "true"^^xsd:boolean ;
1437         :maxSimultaneousStreams 2 ;
1438         :subscriptionCost 9.99 .
1439
1440
1441 ### http://semantics.id/ns/example/film#plan_max_prem
1442 :plan_max_prem rdf:type owl:NamedIndividual ,
1443                 :PremiumPlan ;
1444         :hasAds "false"^^xsd:boolean ;
1445         :maxSimultaneousStreams 4 ;
1446         :subscriptionCost 19.99 .
1447
1448
1449 ### http://semantics.id/ns/example/film#plan_net_ad
1450 :plan_net_ad rdf:type owl:NamedIndividual ,
1451                 :AdSupportedPlan ;
1452         :hasAds "true"^^xsd:boolean ;
1453         :maxSimultaneousStreams 2 ;
1454         :subscriptionCost 6.99 .
1455
1456
1457 ### http://semantics.id/ns/example/film#plan_net_basic
1458 :plan_net_basic rdf:type owl:NamedIndividual ,
1459                 :BasicPlan ;
1460         :hasAds "false"^^xsd:boolean ;
1461         :maxSimultaneousStreams 1 ;
1462         :subscriptionCost 9.99 .
1463
1464
1465 ### http://semantics.id/ns/example/film#plan_net_prem
1466 :plan_net_prem rdf:type owl:NamedIndividual ,
1467                 :PremiumPlan ;
1468         :hasAds "false"^^xsd:boolean ;
1469         :maxSimultaneousStreams 4 ;
1470         :subscriptionCost 15.99 .
1471
1472
1473 ### http://semantics.id/ns/example/film#plan_prime_basic
1474 :plan_prime_basic rdf:type owl:NamedIndividual ,
1475                 :BasicPlan ;
1476         :hasAds "true"^^xsd:boolean ;
1477         :maxSimultaneousStreams 2 ;
1478         :subscriptionCost 8.99 .
1479
1480
1481 ### http://semantics.id/ns/example/film#plan_prime_prem
1482 :plan_prime_prem rdf:type owl:NamedIndividual ,
1483                 :PremiumPlan ;
1484         :hasAds "false"^^xsd:boolean ;
1485         :maxSimultaneousStreams 3 ;
1486         :subscriptionCost 14.99 .

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1487
1488
1489 ### http://semantics.id/ns/example/film#platform_disney
1490 :platform_disney rdf:type owl:NamedIndividual ,
1491                  :StreamingPlatform ;
1492                  :availableInCountry :country_germany ,
1493                                      :country_romania ,
1494                                      :country_uk ,
1495                                      :country_usa ;
1496                  :offersPlan :plan_dis_basic ,
1497                              :plan_dis_fam ;
1498                  :platformURL "www.disneyplus.com" ;
1499                  rdfs:label "Disney+" .
1500
1501
1502 ### http://semantics.id/ns/example/film#platform_hulu
1503 :platform_hulu rdf:type owl:NamedIndividual ,
1504                  :StreamingPlatform ;
1505                  :availableInCountry :country_japan ,
1506                                      :country_usa ;
1507                  :offersPlan :plan_hulu_ad ,
1508                              :plan_hulu_basic ,
1509                              :plan_hulu_stu ;
1510                  :platformURL "www.hulu.com" ;
1511                  rdfs:label "Hulu" .
1512
1513
1514 ### http://semantics.id/ns/example/film#platform_max
1515 :platform_max rdf:type owl:NamedIndividual ,
1516                :StreamingPlatform ;
1517                :availableInCountry :country_germany ,
1518                                    :country_romania ,
1519                                    :country_spain ,
1520                                    :country_usa ;
1521                :offersPlan :plan_max_ad ,
1522                            :plan_max_prem ;
1523                :platformURL "www.max.com" ;
1524                rdfs:label "Max" .
1525
1526
1527 ### http://semantics.id/ns/example/film#platform_netflix
1528 :platform_netflix rdf:type owl:NamedIndividual ,
1529                    :StreamingPlatform ;
1530                    :availableInCountry :country_romania ,
1531                                        :country_spain ,
1532                                        :country_uk ,
1533                                        :country_usa ;
1534                    :offersPlan :plan_net_ad ,
1535                                :plan_net_basic ,
1536                                :plan_net_prem ;
1537                    :platformURL "www.netflix.com" ;
1538                    rdfs:label "Netflix" .
1539
1540
1541 ### http://semantics.id/ns/example/film#platform_prime
1542 :platform_prime rdf:type owl:NamedIndividual ,
1543                  :StreamingPlatform ;
1544                  :availableInCountry :country_romania ,
1545                                      :country_spain ,
1546                                      :country_uk ,
1547                                      :country_usa ;
1548                  :offersPlan :plan_prime_basic ,
1549                              :plan_prime_prem ;
1550                  :platformURL "www.amazon.com/video" ;
1551                  rdfs:label "Prime Video" .
1552
1553
1554 ### http://semantics.id/ns/example/film#rev_alex_1
1555 :rev_alex_1 rdf:type owl:NamedIndividual ,
1556              :Review ;
1557              :reviewsArtwork :movie_dune_2 ;
1558              :reviewScore 9 .
1559
1560
1561 ### http://semantics.id/ns/example/film#rev_alex_2
1562 :rev_alex_2 rdf:type owl:NamedIndividual ,

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1563         :Review ;
1564         :reviewsArtwork :series_chernobyl ;
1565         :reviewScore 9 .
1566
1567
1568 ### http://semantics.id/ns/example/film#rev_alex_3
1569 :rev_alex_3 rdf:type owl:NamedIndividual ,
1570             :Review ;
1571             :reviewsArtwork :movie_interstellar ;
1572             :reviewScore 10 .
1573
1574
1575 ### http://semantics.id/ns/example/film#rev_and_1
1576 :rev_and_1 rdf:type owl:NamedIndividual ,
1577            :Review ;
1578            :reviewsArtwork :movie_collectiv ;
1579            :reviewScore 9 .
1580
1581
1582 ### http://semantics.id/ns/example/film#rev_and_2
1583 :rev_and_2 rdf:type owl:NamedIndividual ,
1584            :Review ;
1585            :reviewsArtwork :series_chernobyl ;
1586            :reviewScore 10 .
1587
1588
1589 ### http://semantics.id/ns/example/film#rev_and_3
1590 :rev_and_3 rdf:type owl:NamedIndividual ,
1591            :Review ;
1592            :reviewsArtwork :movie_dune_2 ;
1593            :reviewScore 9 .
1594
1595
1596 ### http://semantics.id/ns/example/film#rev_car_1
1597 :rev_car_1 rdf:type owl:NamedIndividual ,
1598            :Review ;
1599            :reviewsArtwork :movie_inception ;
1600            :reviewScore 9 .
1601
1602
1603 ### http://semantics.id/ns/example/film#rev_car_2
1604 :rev_car_2 rdf:type owl:NamedIndividual ,
1605            :Review ;
1606            :reviewsArtwork :movie_dark_knight ;
1607            :reviewScore 9 .
1608
1609
1610 ### http://semantics.id/ns/example/film#rev_dav_1
1611 :rev_dav_1 rdf:type owl:NamedIndividual ,
1612            :Review ;
1613            :reviewsArtwork :movie_inception ;
1614            :reviewScore 8 .
1615
1616
1617 ### http://semantics.id/ns/example/film#rev_dav_2
1618 :rev_dav_2 rdf:type owl:NamedIndividual ,
1619            :Review ;
1620            :reviewsArtwork :movie_matrix ;
1621            :reviewScore 9 .
1622
1623
1624 ### http://semantics.id/ns/example/film#rev_dav_3
1625 :rev_dav_3 rdf:type owl:NamedIndividual ,
1626            :Review ;
1627            :reviewsArtwork :movie_dark_knight ;
1628            :reviewScore 10 .
1629
1630
1631 ### http://semantics.id/ns/example/film#rev_dav_4
1632 :rev_dav_4 rdf:type owl:NamedIndividual ,
1633            :Review ;
1634            :reviewsArtwork :movie_parasite ;
1635            :reviewScore 10 .
1636
1637
1638 ### http://semantics.id/ns/example/film#rev_dav_5

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1639 :rev_dav_5 rdf:type owl:NamedIndividual ,
1640         :Review ;
1641         :reviewsArtwork :series_chernobyl ;
1642         :reviewScore 10 .
1643
1644
1645 ### http://semantics.id/ns/example/film#rev_emm_1
1646 :rev_emm_1 rdf:type owl:NamedIndividual ,
1647         :Review ;
1648         :reviewsArtwork :movie_interstellar ;
1649         :reviewScore 9 .
1650
1651
1652 ### http://semantics.id/ns/example/film#rev_emm_2
1653 :rev_emm_2 rdf:type owl:NamedIndividual ,
1654         :Review ;
1655         :reviewsArtwork :movie_dune_2 ;
1656         :reviewScore 10 .
1657
1658
1659 ### http://semantics.id/ns/example/film#rev_emm_3
1660 :rev_emm_3 rdf:type owl:NamedIndividual ,
1661         :Review ;
1662         :reviewsArtwork :movie_barbie ;
1663         :reviewScore 7 .
1664
1665
1666 ### http://semantics.id/ns/example/film#rev_isa_1
1667 :rev_isa_1 rdf:type owl:NamedIndividual ,
1668         :Review ;
1669         :reviewsArtwork :movie_parasite ;
1670         :reviewScore 10 .
1671
1672
1673 ### http://semantics.id/ns/example/film#rev_isa_2
1674 :rev_isa_2 rdf:type owl:NamedIndividual ,
1675         :Review ;
1676         :reviewsArtwork :movie_gladiator ;
1677         :reviewScore 7 .
1678
1679
1680 ### http://semantics.id/ns/example/film#rev_isa_3
1681 :rev_isa_3 rdf:type owl:NamedIndividual ,
1682         :Review ;
1683         :reviewsArtwork :movie_dark_knight ;
1684         :reviewScore 9 .
1685
1686
1687 ### http://semantics.id/ns/example/film#rev_ken_1
1688 :rev_ken_1 rdf:type owl:NamedIndividual ,
1689         :Review ;
1690         :reviewsArtwork :movie_spirited ;
1691         :reviewScore 10 .
1692
1693
1694 ### http://semantics.id/ns/example/film#rev_ken_2
1695 :rev_ken_2 rdf:type owl:NamedIndividual ,
1696         :Review ;
1697         :reviewsArtwork :movie_parasite ;
1698         :reviewScore 9 .
1699
1700
1701 ### http://semantics.id/ns/example/film#rev_ken_3
1702 :rev_ken_3 rdf:type owl:NamedIndividual ,
1703         :Review ;
1704         :reviewsArtwork :series_chernobyl ;
1705         :reviewScore 10 .
1706
1707
1708 ### http://semantics.id/ns/example/film#rev_luc_1
1709 :rev_luc_1 rdf:type owl:NamedIndividual ,
1710         :Review ;
1711         :reviewsArtwork :movie_oppenheimer ;
1712         :reviewScore 9 .
1713
1714

```

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1715 ### http://semantics.id/ns/example/film#rev_luc_2
1716 :rev_luc_2 rdf:type owl:NamedIndividual ,
1717             :Review ;
1718             :reviewsArtwork :movie_dune_2 ;
1719             :reviewScore 8 .
1720
1721
1722 ### http://semantics.id/ns/example/film#rev_luc_3
1723 :rev_luc_3 rdf:type owl:NamedIndividual ,
1724             :Review ;
1725             :reviewsArtwork :series_bcs ;
1726             :reviewScore 10 .
1727
1728
1729 ### http://semantics.id/ns/example/film#rev_mar_1
1730 :rev_mar_1 rdf:type owl:NamedIndividual ,
1731             :Review ;
1732             :reviewsArtwork :series_mh ;
1733             :reviewScore 10 .
1734
1735
1736 ### http://semantics.id/ns/example/film#rev_mar_2
1737 :rev_mar_2 rdf:type owl:NamedIndividual ,
1738             :Review ;
1739             :reviewsArtwork :movie_barbie ;
1740             :reviewScore 8 .
1741
1742
1743 ### http://semantics.id/ns/example/film#rev_mar_3
1744 :rev_mar_3 rdf:type owl:NamedIndividual ,
1745             :Review ;
1746             :reviewsArtwork :series_bcs ;
1747             :reviewScore 10 .
1748
1749
1750 ### http://semantics.id/ns/example/film#rev_mat_1
1751 :rev_mat_1 rdf:type owl:NamedIndividual ,
1752             :Review ;
1753             :reviewsArtwork :movie_432 ;
1754             :reviewScore 10 .
1755
1756
1757 ### http://semantics.id/ns/example/film#rev_mat_2
1758 :rev_mat_2 rdf:type owl:NamedIndividual ,
1759             :Review ;
1760             :reviewsArtwork :movie_oppenheimer ;
1761             :reviewScore 9 .
1762
1763
1764 ### http://semantics.id/ns/example/film#rev_mat_3
1765 :rev_mat_3 rdf:type owl:NamedIndividual ,
1766             :Review ;
1767             :reviewsArtwork :movie_gladiator ;
1768             :reviewScore 8 .
1769
1770
1771 ### http://semantics.id/ns/example/film#rev_mat_4
1772 :rev_mat_4 rdf:type owl:NamedIndividual ,
1773             :Review ;
1774             :reviewsArtwork :series_arcane ;
1775             :reviewScore 10 .
1776
1777
1778 ### http://semantics.id/ns/example/film#rev_sop_1
1779 :rev_sop_1 rdf:type owl:NamedIndividual ,
1780             :Review ;
1781             :reviewsArtwork :series_chernobyl ;
1782             :reviewScore 10 .
1783
1784
1785 ### http://semantics.id/ns/example/film#rev_sop_2
1786 :rev_sop_2 rdf:type owl:NamedIndividual ,
1787             :Review ;
1788             :reviewsArtwork :movie_spirited ;
1789             :reviewScore 10 .
1790

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1791 :movie_matrix :sequel :movie_dark_knight .
1792 :movie_dark_knight :sequel :movie_inception .
1793 :movie_inception :inSameUniverse :movie_interstellar .
1794 :user_alex :recommendsTo :user_matei .
1795
1796 ### http://semantics.id/ns/example/film#rev_sop_3
1797 :rev_sop_3 rdf:type owl:NamedIndividual ,
1798             :Review ;
1799             :reviewsArtwork :series_mh ;
1800             :reviewScore 9 .
1801
1802
1803 ### http://semantics.id/ns/example/film#rev_sop_4
1804 :rev_sop_4 rdf:type owl:NamedIndividual ,
1805             :Review ;
1806             :reviewsArtwork :movie_dark_knight ;
1807             :reviewScore 8 .
1808
1809
1810 ### http://semantics.id/ns/example/film#season_arc_1
1811 :season_arc_1 rdf:type owl:NamedIndividual ,
1812               :Season ;
1813               :hasEpisode :ep_arc_1 ,
1814                           :ep_arc_2 ,
1815                           :ep_arc_3 ,
1816                           :ep_arc_4 ,
1817                           :ep_arc_5 ,
1818                           :ep_arc_6 ,
1819                           :ep_arc_7 ,
1820                           :ep_arc_8 ,
1821                           :ep_arc_9 ;
1822               rdfs:label "Arcane: Season 1" .
1823
1824
1825 ### http://semantics.id/ns/example/film#season_bcs_1
1826 :season_bcs_1 rdf:type owl:NamedIndividual ,
1827               :Season ;
1828               :hasEpisode :ep_bcs_101 ,
1829                           :ep_bcs_102 ,
1830                           :ep_bcs_103 ,
1831                           :ep_bcs_104 ,
1832                           :ep_bcs_105 ,
1833                           :ep_bcs_106 ,
1834                           :ep_bcs_107 ,
1835                           :ep_bcs_108 ,
1836                           :ep_bcs_109 ,
1837                           :ep_bcs_110 ;
1838               rdfs:label "BCS: Season 1" .
1839
1840
1841 ### http://semantics.id/ns/example/film#season_chernobyl_1
1842 :season_chernobyl_1 rdf:type owl:NamedIndividual ,
1843                       :Season ;
1844                       :hasEpisode :ep_cher_1 ,
1845                                   :ep_cher_2 ,
1846                                   :ep_cher_3 ,
1847                                   :ep_cher_4 ,
1848                                   :ep_cher_5 ;
1849                       rdfs:label "Chernobyl: Season 1" .
1850
1851
1852 ### http://semantics.id/ns/example/film#season_mh_1
1853 :season_mh_1 rdf:type owl:NamedIndividual ,
1854              :Season ;
1855              :hasEpisode :ep_mh_1 ,
1856                          :ep_mh_2 ,
1857                          :ep_mh_3 ,
1858                          :ep_mh_4 ,
1859                          :ep_mh_5 ;
1860              rdfs:label "Money Heist: Part 1" .
1861
1862
1863 ### http://semantics.id/ns/example/film#series_arcane
1864 :series_arcane rdf:type owl:NamedIndividual ,
1865                :Series ;
1866                :hasGenre :genre_animation ;

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1867         :hasOriginCountry :country_usa ;
1868         :hasSeason :season_arc_1 ;
1869         rdfs:label "Arcane" .
1870
1871
1872 ### http://semantics.id/ns/example/film#series_bcs
1873 :series_bcs rdf:type owl:NamedIndividual ,
1874             :Series ;
1875         :hasGenre :genre_drama ;
1876         :hasOriginCountry :country_usa ;
1877         :hasSeason :season_bcs_1 ;
1878         rdfs:label "Better Call Saul" .
1879
1880
1881 ### http://semantics.id/ns/example/film#series_chernobyl
1882 :series_chernobyl rdf:type owl:NamedIndividual ,
1883                     :Miniseries ;
1884         :hasGenre :genre_drama ;
1885         :hasOriginCountry :country_uk ;
1886         :hasSeason :season_chernobyl_1 ;
1887         rdfs:label "Chernobyl" .
1888
1889
1890 ### http://semantics.id/ns/example/film#series_mh
1891 :series_mh rdf:type owl:NamedIndividual ,
1892            :Series ;
1893         :hasGenre :genre_action ;
1894         :hasOriginCountry :country_spain ;
1895         :hasSeason :season_mh_1 ;
1896         rdfs:label "Money Heist" .
1897
1898
1899 ### http://semantics.id/ns/example/film#user_alex
1900 :user_alex rdf:type owl:NamedIndividual ,
1901             :Subscriber ;
1902         :friendOf :user_andrei ,
1903                 :user_maria ,
1904                 :user_matei ;
1905         :hasPlan :plan_max_prem ,
1906                 :plan_net_prem ;
1907         :likes :movie_dune_2 ,
1908               :movie_interstellar ,
1909               :series_chernobyl ;
1910         :locatedIn :country_romania ;
1911         :subscribedTo :platform_max ,
1912                     :platform_netflix ;
1913         :watched :movie_dune_2 ,
1914                 :movie_interstellar ,
1915                 :movie_oppenheimer ,
1916                 :movie_parasite ,
1917                 :series_chernobyl ;
1918         :wroteReview :rev_alex_1 ,
1919                    :rev_alex_2 ,
1920                    :rev_alex_3 ;
1921         :subscriberAge 22 ;
1922         rdfs:label "Alex" .
1923
1924
1925 ### http://semantics.id/ns/example/film#user_andrei
1926 :user_andrei rdf:type owl:NamedIndividual ,
1927               :Subscriber ;
1928         :friendOf :user_alex ,
1929                 :user_matei ;
1930         :hasPlan :plan_max_prem ;
1931         :likes :movie_colectiv ,
1932               :movie_dune_2 ,
1933               :series_chernobyl ;
1934         :locatedIn :country_romania ;
1935         :subscribedTo :platform_max ;
1936         :watched :movie_barbie ,
1937                 :movie_colectiv ,
1938                 :movie_dune_2 ,
1939                 :movie_matrix ,
1940                 :series_chernobyl ;
1941         :wroteReview :rev_and_1 ,
1942                    :rev_and_2 ,

```



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1943         :rev_and_3 ;
1944     :subscriberAge 28 ;
1945     rdfs:label "Andrei" .
1946
1947
1948 ### http://semantics.id/ns/example/film#user_carmen
1949 :user_carmen rdf:type owl:NamedIndividual ,
1950             :Subscriber ;
1951     :friendOf :user_isa ,
1952             :user_maria ;
1953     :hasPlan :plan_net_ad ;
1954     :likes :movie_dark_knight ,
1955           :movie_inception ,
1956           :series_mh ;
1957     :locatedIn :country_spain ;
1958     :subscribedTo :platform_netflix ;
1959     :watched :movie_dark_knight ,
1960             :movie_gladiator ,
1961             :movie_inception ,
1962             :series_bcs ,
1963             :series_mh ;
1964     :wroteReview :rev_car_1 ,
1965                :rev_car_2 ;
1966     :subscriberAge 26 ;
1967     rdfs:label "Carmen" .
1968
1969
1970 ### http://semantics.id/ns/example/film#user_david
1971 :user_david rdf:type owl:NamedIndividual ,
1972             :Subscriber ;
1973     :friendOf :user_alex ,
1974             :user_emma ,
1975             :user_lucas ;
1976     :hasPlan :plan_hulu_ad ,
1977             :plan_max_ad ;
1978     :likes :movie_dark_knight ,
1979           :movie_inception ,
1980           :movie_matrix ,
1981           :movie_parasite ,
1982           :series_chernobyl ;
1983     :locatedIn :country_usa ;
1984     :subscribedTo :platform_hulu ,
1985                 :platform_max ;
1986     :watched :movie_dark_knight ,
1987             :movie_inception ,
1988             :movie_matrix ,
1989             :movie_parasite ,
1990             :series_chernobyl ;
1991     :wroteReview :rev_dav_1 ,
1992                :rev_dav_2 ,
1993                :rev_dav_3 ,
1994                :rev_dav_4 ,
1995                :rev_dav_5 ;
1996     :subscriberAge 21 ;
1997     rdfs:label "David" .
1998
1999
2000 ### http://semantics.id/ns/example/film#user_emma
2001 :user_emma rdf:type owl:NamedIndividual ,
2002             :Subscriber ;
2003     :friendOf :user_david ,
2004             :user_lucas ,
2005             :user_sophia ;
2006     :hasPlan :plan_prime_basic ;
2007     :likes :movie_barbie ,
2008           :movie_dune_2 ,
2009           :movie_interstellar ,
2010           :series_bcs ;
2011     :locatedIn :country_uk ;
2012     :subscribedTo :platform_prime ;
2013     :watched :movie_barbie ,
2014             :movie_dune_2 ,
2015             :movie_interstellar ,
2016             :movie_matrix ,
2017             :series_bcs ;
2018     :wroteReview :rev_emm_1 ,

```

```

2019         :rev_emm_2 ,
2020         :rev_emm_3 ;
2021     :subscriberAge 26 ;
2022     rdfs:label "Emma" .
2023
2024
2025 ### http://semantics.id/ns/example/film#user_isa
2026 :user_isa rdf:type owl:NamedIndividual ,
2027         :Subscriber ;
2028     :friendOf :user_carmen ,
2029         :user_maria ;
2030     :hasPlan :plan_max_prem ;
2031     :likes :movie_dark_knight ,
2032         :movie_gladiator ,
2033         :movie_parasite ,
2034         :series_arcane ;
2035     :locatedIn :country_spain ;
2036     :subscribedTo :platform_max ;
2037     :watched :movie_dark_knight ,
2038         :movie_gladiator ,
2039         :movie_oppenheimer ,
2040         :movie_parasite ,
2041         :series_arcane ;
2042     :wroteReview :rev_isa_1 ,
2043         :rev_isa_2 ,
2044         :rev_isa_3 ;
2045     :subscriberAge 23 ;
2046     rdfs:label "Isabella" .
2047
2048
2049 ### http://semantics.id/ns/example/film#user_ken
2050 :user_ken rdf:type owl:NamedIndividual ,
2051         :Subscriber ;
2052     :friendOf :user_david ,
2053         :user_sophia ;
2054     :hasPlan :plan_hulu_stu ;
2055     :likes :movie_parasite ,
2056         :movie_spirited ,
2057         :series_chernobyl ;
2058     :locatedIn :country_japan ;
2059     :subscribedTo :platform_hulu ;
2060     :watched :movie_interstellar ,
2061         :movie_parasite ,
2062         :movie_spirited ,
2063         :series_arcane ,
2064         :series_chernobyl ;
2065     :wroteReview :rev_ken_1 ,
2066         :rev_ken_2 ,
2067         :rev_ken_3 ;
2068     :subscriberAge 30 ;
2069     rdfs:label "Ken" .
2070
2071
2072 ### http://semantics.id/ns/example/film#user_lucas
2073 :user_lucas rdf:type owl:NamedIndividual ,
2074         :Subscriber ;
2075     :friendOf :user_david ,
2076         :user_emma ;
2077     :hasPlan :plan_net_basic ;
2078     :likes :movie_dune_2 ,
2079         :movie_oppenheimer ,
2080         :series_bcs ;
2081     :locatedIn :country_usa ;
2082     :subscribedTo :platform_netflix ;
2083     :watched :movie_dune_2 ,
2084         :movie_interstellar ,
2085         :movie_matrix ,
2086         :movie_oppenheimer ,
2087         :series_bcs ;
2088     :wroteReview :rev_luc_1 ,
2089         :rev_luc_2 ,
2090         :rev_luc_3 ;
2091     :subscriberAge 28 ;
2092     rdfs:label "Lucas" .
2093
2094

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2095 ### http://semantics.id/ns/example/film#user_maria
2096 :user_maria rdf:type owl:NamedIndividual ,
2097             :Subscriber ;
2098             :friendOf :user_carmen ,
2099                     :user_david ,
2100                     :user_isa ;
2101             :hasPlan :plan_dis_basic ,
2102                     :plan_net_ad ;
2103             :likes :movie_barbie ,
2104                     :series_bcs ,
2105                     :series_chernobyl ,
2106                     :series_mh ;
2107             :locatedIn :country_spain ;
2108             :subscribedTo :platform_disney ,
2109                     :platform_netflix ;
2110             :watched :movie_barbie ,
2111                     :movie_inception ,
2112                     :series_bcs ,
2113                     :series_chernobyl ,
2114                     :series_mh ;
2115             :wroteReview :rev_mar_1 ,
2116                     :rev_mar_2 ,
2117                     :rev_mar_3 ;
2118             :subscriberAge 24 ;
2119             rdfs:label "Maria" .
2120
2121
2122 ### http://semantics.id/ns/example/film#user_matei
2123 :user_matei rdf:type owl:NamedIndividual ,
2124             :Subscriber ;
2125             :friendOf :user_alex ,
2126                     :user_andrei ;
2127             :hasPlan :plan_max_prem ;
2128             :likes :movie_432 ,
2129                     :movie_gladiator ,
2130                     :movie_oppenheimer ,
2131                     :series_arcane ;
2132             :locatedIn :country_romania ;
2133             :subscribedTo :platform_max ;
2134             :watched :movie_432 ,
2135                     :movie_chuck ,
2136                     :movie_gladiator ,
2137                     :movie_oppenheimer ,
2138                     :series_arcane ;
2139             :wroteReview :rev_mat_1 ,
2140                     :rev_mat_2 ,
2141                     :rev_mat_3 ,
2142                     :rev_mat_4 ;
2143             :subscriberAge 25 ;
2144             rdfs:label "Matei" .
2145
2146
2147 ### http://semantics.id/ns/example/film#user_sophia
2148 :user_sophia rdf:type owl:NamedIndividual ,
2149             :Subscriber ;
2150             :friendOf :user_emma ,
2151                     :user_ken ,
2152                     :user_lucas ;
2153             :hasPlan :plan_dis_fam ;
2154             :likes :movie_dark_knight ,
2155                     :movie_spirited ,
2156                     :series_arcane ,
2157                     :series_chernobyl ,
2158                     :series_mh ;
2159             :locatedIn :country_germany ;
2160             :subscribedTo :platform_disney ;
2161             :watched :movie_dark_knight ,
2162                     :movie_spirited ,
2163                     :series_arcane ,
2164                     :series_chernobyl ,
2165                     :series_mh ;
2166             :wroteReview :rev_sop_1 ,
2167                     :rev_sop_2 ,
2168                     :rev_sop_3 ,
2169                     :rev_sop_4 ;
2170             :subscriberAge 25 ;

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```

2171         rdfs:label "Sophia" .
2172
2173
2174 ### http://semantics.id/ns/example/film#zendaya
2175 :zendaya rdf:type owl:NamedIndividual ,
2176           :Actor ;
2177           :dateOfBirth "1996-09-01"^^xsd:date ;
2178           :fullName "Zendaya Maree Stoermer Coleman" ;
2179           :gender "female" .
2180
2181
2182 #####
2183 # Rules
2184 #####
2185
2186 :s rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2187
2188 :e1 rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2189
2190 :season rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2191
2192 :e2 rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2193
2194 :f rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2195
2196 :a1 rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2197
2198 :g rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2199
2200 :film rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2201
2202 :r rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2203
2204 :score rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2205
2206 :p rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2207
2208 :platform rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2209
2210 :c rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2211
2212 :series rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2213
2214 :a rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2215
2216 :b rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2217
2218 :x rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2219
2220 :a2 rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2221
2222 :age rdf:type <http://www.w3.org/2003/11/swrl#Variable> .
2223
2224 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2225   rdfs:comment "If subscriber A is a friend of B, and B likes artwork X, and A has not watched X, then
2226     ↪ recommend X to A." ;
2227   rdfs:label "recommendToRule" ;
2228   rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2229   <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2230     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2231       <http://www.w3.org/2003/11/swrl#classPredicate> :
2232         ↪ Subscriber ;
2233       <http://www.w3.org/2003/11/swrl#argument1> :a
2234     ] ;
2235     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2236       rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2237         ↪ IndividualPropertyAtom> ;
2238         <http://www.w3.org/2003/11/swrl#
2239         ↪ propertyPredicate> :friendOf ;
2240         <http://www.w3.org/2003/11/swrl#argument1> :
2241         ↪ a ;
2242         <http://www.w3.org/2003/11/swrl#argument2> :
2243         ↪ b
2244       ] ;
2245     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
2246       ↪ AtomList> ;

```

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2240                                     rdf:first [ rdf:type <http://www.w3.org
↳ /2003/11/swrl#IndividualPropertyAtom> ;
2241                                     <http://www.w3.org/2003/11/swrl#
↳ propertyPredicate> :likes ;
2242                                     <http://www.w3.org/2003/11/swrl#
↳ argument1> :b ;
2243                                     <http://www.w3.org/2003/11/swrl#
↳ argument2> :x
2244                                     ] ;
2245                                     rdf:rest rdf:nil
2246                                     ]
2247                                 ]
2248                                 ] ;
2249 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2250                                     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ IndividualPropertyAtom> ;
2251                                     <http://www.w3.org/2003/11/swrl#propertyPredicate> :
↳ recommendsTo ;
2252                                     <http://www.w3.org/2003/11/swrl#argument1> :x ;
2253                                     <http://www.w3.org/2003/11/swrl#argument2> :a
2254                                     ] ;
2255                                     rdf:rest rdf:nil
2256                                 ]
2257 ] .
2258
2259 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2260   rdfs:comment "If a subscriber likes one artwork, and another artwork has the same genre, recommend that
↳ other artwork." ;
2261   rdfs:label "recommendsToGenreRule" ;
2262   rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2263   <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2264                                           rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2265                                           <http://www.w3.org/2003/11/swrl#classPredicate> :
↳ Subscriber ;
2266                                           <http://www.w3.org/2003/11/swrl#argument1> :s
2267                                           ] ;
2268                                           rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2269                                           rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ IndividualPropertyAtom> ;
2270                                           <http://www.w3.org/2003/11/swrl#
↳ propertyPredicate> :likes ;
2271                                           <http://www.w3.org/2003/11/swrl#argument1> :
↳ s ;
2272                                           <http://www.w3.org/2003/11/swrl#argument2> :
↳ a1
2273                                           ] ;
2274                                           rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ AtomList> ;
2275                                           rdf:first [ rdf:type <http://www.w3.org
↳ /2003/11/swrl#IndividualPropertyAtom> ;
2276                                           <http://www.w3.org/2003/11/swrl#
↳ propertyPredicate> :hasGenre ;
2277                                           <http://www.w3.org/2003/11/swrl#
↳ argument1> :a1 ;
2278                                           <http://www.w3.org/2003/11/swrl#
↳ argument2> :g
2279                                           ] ;
2280                                           rdf:rest [ rdf:type <http://www.w3.org
↳ /2003/11/swrl#AtomList> ;
2281                                           rdf:first [ rdf:type <http://www.
↳ w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2282                                           <http://www.w3.org
↳ /2003/11/swrl#propertyPredicate> :hasGenre ;
2283                                           <http://www.w3.org
↳ /2003/11/swrl#argument1> :a2 ;
2284                                           <http://www.w3.org
↳ /2003/11/swrl#argument2> :g
2285                                           ] ;
2286                                           rdf:rest rdf:nil
2287                                           ]
2288                                           ]
2289                                           ]
2290                                           ] ;
2291 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2292                                     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ IndividualPropertyAtom> ;

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2293         <http://www.w3.org/2003/11/swrl#propertyPredicate> :
2294         ↪ recommendsTo ;
2295         <http://www.w3.org/2003/11/swrl#argument1> :a2 ;
2296         <http://www.w3.org/2003/11/swrl#argument2> :s
2297     ] ;
2298     rdf:rest rdf:nil
2299 ] .
2300
2301 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2302   rdfs:comment "If a subscriber has an age greater than 25, classify them as an AdultSubscriber." ;
2303   rdfs:label "AdultSubscriberRule" ;
2304   rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2305   <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2306     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2307       <http://www.w3.org/2003/11/swrl#classPredicate> :
2308         ↪ Subscriber ;
2309       <http://www.w3.org/2003/11/swrl#argument1> :s
2310     ] ;
2311     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2312       rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2313         ↪ DatavaluedPropertyAtom> ;
2314         <http://www.w3.org/2003/11/swrl#
2315         ↪ propertyPredicate> :subscriberAge ;
2316         <http://www.w3.org/2003/11/swrl#argument1> :
2317         ↪ s ;
2318         <http://www.w3.org/2003/11/swrl#argument2> :
2319         ↪ age
2320       ] ;
2321       rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
2322         ↪ AtomList> ;
2323         rdf:first [ rdf:type <http://www.w3.org
2324         ↪ /2003/11/swrl#BuiltinAtom> ;
2325         <http://www.w3.org/2003/11/swrl#
2326         ↪ builtin> <http://www.w3.org/2003/11/swrlb#greaterThan> ;
2327         <http://www.w3.org/2003/11/swrl#
2328         ↪ arguments> [ rdf:type rdf:List ;
2329         ↪
2330         ↪     rdf:first :age ;
2331         ↪
2332         ↪     rdf:rest [ rdf:type rdf:List ;
2333         ↪
2334         ↪         rdf:first 25 ;
2335         ↪
2336         ↪         rdf:rest rdf:nil
2337       ]
2338     ]
2339   ] ;
2340   <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2341     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2342       <http://www.w3.org/2003/11/swrl#classPredicate> :
2343         ↪ AdultSubscriber ;
2344       <http://www.w3.org/2003/11/swrl#argument1> :s
2345     ] ;
2346     rdf:rest rdf:nil
2347   ]
2348 ] .
2349
2350 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2351   rdfs:comment "If a subscriber watched an episode, and that episode belongs to a season that contains other
2352     ↪ episodes, mark those other episodes as being in the subscriber's series-in-progress." ;
2353   rdfs:label "inProgressRule" ;
2354   rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2355   <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2356     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2357       <http://www.w3.org/2003/11/swrl#classPredicate> :
2358         ↪ Subscriber ;
2359       <http://www.w3.org/2003/11/swrl#argument1> :s
2360     ] ;
2361     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;

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2350         rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ IndividualPropertyAtom> ;
2351         <http://www.w3.org/2003/11/swrl#
↳ propertyPredicate> :watched ;
2352         <http://www.w3.org/2003/11/swrl#argument1> :
↳ s ;
2353         <http://www.w3.org/2003/11/swrl#argument2> :
↳ e1
2354     ] ;
2355     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ AtomList> ;
2356         rdf:first [ rdf:type <http://www.w3.org
↳ /2003/11/swrl#ClassAtom> ;
2357         <http://www.w3.org/2003/11/swrl#
↳ classPredicate> :Episode ;
2358         <http://www.w3.org/2003/11/swrl#
↳ argument1> :e1
2359     ] ;
2360     rdf:rest [ rdf:type <http://www.w3.org
↳ /2003/11/swrl#AtomList> ;
2361         rdf:first [ rdf:type <http://www.
↳ w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2362         <http://www.w3.org
↳ /2003/11/swrl#propertyPredicate> :hasEpisode ;
2363         <http://www.w3.org
↳ /2003/11/swrl#argument1> :season ;
2364         <http://www.w3.org
↳ /2003/11/swrl#argument2> :e1
2365     ] ;
2366     rdf:rest [ rdf:type <http://www.w3
↳ .org/2003/11/swrl#AtomList> ;
2367         rdf:first [ rdf:type <
↳ http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2368         <http://www
↳ .w3.org/2003/11/swrl#propertyPredicate> :hasEpisode ;
2369         <http://www
↳ .w3.org/2003/11/swrl#argument1> :season ;
2370         <http://www
↳ .w3.org/2003/11/swrl#argument2> :e2
2371     ] ;
2372     rdf:rest [ rdf:type <
↳ http://www.w3.org/2003/11/swrl#AtomList> ;
2373         rdf:first [
↳ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2374         <http://www.w3.org/2003/11/swrl#classPredicate> :Episode ;
2375         <http://www.w3.org/2003/11/swrl#argument1> :e2
2376     ]
↳ ;
2377     rdf:rest rdf:nil
2378 ]
2379 ]
2380 ]
2381 ]
2382 ]
2383 ] ;
2384 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2385     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
↳ IndividualPropertyAtom> ;
2386         <http://www.w3.org/2003/11/swrl#propertyPredicate> :
↳ inProgress ;
2387         <http://www.w3.org/2003/11/swrl#argument1> :e2 ;
2388         <http://www.w3.org/2003/11/swrl#argument2> :s
2389     ] ;
2390     rdf:rest rdf:nil
2391 ]
2392 ] .
2393
2394 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2395     rdfs:comment "If a subscription plan has hasAds equal to false, then it's an AdFreePlan." ;
2396     rdfs:label "AdFreePlanRule" ;
2397     rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2398     <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2399         rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;

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2400         <http://www.w3.org/2003/11/swrl#classPredicate> :
    ↪ SubscriptionPlan ;
2401         <http://www.w3.org/2003/11/swrl#argument1> :p
2402     ] ;
2403     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2404         rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
    ↪ DatavaluedPropertyAtom> ;
2405         <http://www.w3.org/2003/11/swrl#
    ↪ propertyPredicate> :hasAds ;
2406         <http://www.w3.org/2003/11/swrl#argument1> :
    ↪ p ;
2407         <http://www.w3.org/2003/11/swrl#argument2> "
    ↪ false"^^xsd:boolean
2408     ] ;
2409     rdf:rest rdf:nil
2410 ]
2411 ] ;
2412 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2413     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2414         <http://www.w3.org/2003/11/swrl#classPredicate> :
    ↪ AdFreePlan ;
2415         <http://www.w3.org/2003/11/swrl#argument1> :p
2416     ] ;
2417     rdf:rest rdf:nil
2418 ]
2419 ] .
2420
2421 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2422     rdfs:comment "If a subscriber's friend likes a film, and a similar-genre film has a review with score >= 9,
    ↪ mark the film as HighlyRecommendedForUser." ;
2423     rdfs:label "HighlyRecommendedForUser" ;
2424     rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2425     <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2426         rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2427             <http://www.w3.org/2003/11/swrl#classPredicate> :
    ↪ Subscriber ;
2428             <http://www.w3.org/2003/11/swrl#argument1> :s
2429         ] ;
2430         rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2431             rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
    ↪ IndividualPropertyAtom> ;
2432             <http://www.w3.org/2003/11/swrl#
    ↪ propertyPredicate> :friendOf ;
2433             <http://www.w3.org/2003/11/swrl#argument1> :
    ↪ s ;
2434             <http://www.w3.org/2003/11/swrl#argument2> :
    ↪ f
2435         ] ;
2436         rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
    ↪ AtomList> ;
2437             rdf:first [ rdf:type <http://www.w3.org
    ↪ /2003/11/swrl#IndividualPropertyAtom> ;
2438                 <http://www.w3.org/2003/11/swrl#
    ↪ propertyPredicate> :likes ;
2439                 <http://www.w3.org/2003/11/swrl#
    ↪ argument1> :f ;
2440                 <http://www.w3.org/2003/11/swrl#
    ↪ argument2> :a1
2441             ] ;
2442             rdf:rest [ rdf:type <http://www.w3.org
    ↪ /2003/11/swrl#AtomList> ;
2443                 rdf:first [ rdf:type <http://www.
    ↪ w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2444                     <http://www.w3.org
    ↪ /2003/11/swrl#propertyPredicate> :hasGenre ;
2445                     <http://www.w3.org
    ↪ /2003/11/swrl#argument1> :a1 ;
2446                     <http://www.w3.org
    ↪ /2003/11/swrl#argument2> :g
2447                 ] ;
2448                 rdf:rest [ rdf:type <http://www.w3
    ↪ .org/2003/11/swrl#AtomList> ;
2449                     rdf:first [ rdf:type <
    ↪ http://www.w3.org/2003/11/swrl#ClassAtom> ;
2450                     <http://www
    ↪ .w3.org/2003/11/swrl#classPredicate> :Film ;

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2451                                     <http://www
↳ .w3.org/2003/11/swrl#argument1> :film
2452                                     ] ;
2453                                     rdf:rest [ rdf:type <
↳ http://www.w3.org/2003/11/swrl#AtomList> ;
2454                                     rdf:first [
↳ rdf:type <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2455                                     <http://www.w3.org/2003/11/swrl#propertyPredicate> :hasGenre ;
2456                                     <http://www.w3.org/2003/11/swrl#argument1> :film ;
2457                                     <http://www.w3.org/2003/11/swrl#argument2> :g
2458                                     ]
↳ ;
2459                                     rdf:rest [
↳ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2460                                     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2461                                     <http://www.w3.org/2003/11/swrl#classPredicate> :Review ;
2462                                     <http://www.w3.org/2003/11/swrl#argument1> :r
2463                                     ] ;
2464                                     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2465                                     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2466                                     <http://www.w3.org/2003/11/swrl#propertyPredicate> :reviewsArtwork ;
2467                                     <http://www.w3.org/2003/11/swrl#argument1> :r ;
2468                                     <http://www.w3.org/2003/11/swrl#argument2> :film
2469                                     ] ;
2470                                     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2471                                     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#DatavaluedPropertyAtom> ;
2472                                     <http://www.w3.org/2003/11/swrl#propertyPredicate> :reviewScore ;
2473                                     <http://www.w3.org/2003/11/swrl#argument1> :r ;
2474                                     <http://www.w3.org/2003/11/swrl#argument2> :score
2475                                     ] ;
2476                                     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2477                                     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#BuiltinAtom> ;
2478                                     <http://www.w3.org/2003/11/swrl#builtin> <http://www.w3.
↳ org/2003/11/swrlb#greaterThanOrEqual> ;
2479                                     <http://www.w3.org/2003/11/swrl#arguments> [ rdf:type rdf:
↳ List ;
2480                                     rdf:first :
↳ score ;
2481                                     rdf:rest [
↳ rdf:type rdf:List ;
2482                                     rdf:first 9 ;
2483                                     rdf:rest rdf:nil
2484                                     ]
↳
2485                                     ]
2486

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2487         ] ;
2488         rdf:rest rdf:nil
2489     ]
2490 ]
2491 ]
2492 ]
2493 ]
2494 ]
2495 ]
2496 ]
2497 ] ;
2498 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2499     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2500         ↪ IndividualPropertyAtom> ;
2501         ↪ recommendsTo ;
2502         ↪ <http://www.w3.org/2003/11/swrl#propertyPredicate> :
2503         ↪ <http://www.w3.org/2003/11/swrl#argument1> :film ;
2504         ↪ <http://www.w3.org/2003/11/swrl#argument2> :s
2505         ] ;
2506         ↪ rdf:rest rdf:nil
2507     ] .
2508 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2509     rdfs:comment "If a subscriber has a premium plan, is located in country C, and their platform is available
2510     ↪ in C, then they have full access." ;
2511     rdfs:label "FullAccessSubscriberRule" ;
2512     rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2513     <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2514         ↪ rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2515             ↪ <http://www.w3.org/2003/11/swrl#classPredicate> :
2516             ↪ <http://www.w3.org/2003/11/swrl#argument1> :s
2517             ] ;
2518             ↪ rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2519                 ↪ rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2520                     ↪ IndividualPropertyAtom> ;
2521                     ↪ <http://www.w3.org/2003/11/swrl#
2522                     ↪ propertyPredicate> :hasPlan ;
2523                     ↪ <http://www.w3.org/2003/11/swrl#argument1> :
2524                     ↪ s ;
2525                     ↪ <http://www.w3.org/2003/11/swrl#argument2> :
2526                     ↪ p
2527                     ] ;
2528                     ↪ rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
2529                         ↪ AtomList> ;
2530                         ↪ rdf:first [ rdf:type <http://www.w3.org
2531                             ↪ /2003/11/swrl#ClassAtom> ;
2532                             ↪ <http://www.w3.org/2003/11/swrl#
2533                             ↪ classPredicate> :PremiumPlan ;
2534                             ↪ <http://www.w3.org/2003/11/swrl#
2535                             ↪ argument1> :p
2536                             ] ;
2537                             ↪ rdf:rest [ rdf:type <http://www.w3.org
2538                                 ↪ /2003/11/swrl#AtomList> ;
2539                                 ↪ rdf:first [ rdf:type <http://www.
2540                                     ↪ w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2541                                     ↪ <http://www.w3.org
2542                                     ↪ /2003/11/swrl#propertyPredicate> :subscribedTo ;
2543                                     ↪ <http://www.w3.org
2544                                     ↪ /2003/11/swrl#argument1> :s ;
2545                                     ↪ <http://www.w3.org
2546                                     ↪ /2003/11/swrl#argument2> :platform
2547                                     ] ;
2548                                     ↪ rdf:rest [ rdf:type <http://www.w3
2549                                         ↪ .org/2003/11/swrl#AtomList> ;
2550                                         ↪ rdf:first [ rdf:type <
2551                                             ↪ http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2552                                             ↪ <http://www
2553                                             ↪ .w3.org/2003/11/swrl#propertyPredicate> :offersPlan ;
2554                                             ↪ <http://www

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2538 ↪ .w3.org/2003/11/swrl#argument1> :platform ;
2539
2540 ↪ .w3.org/2003/11/swrl#argument2> :p
2541
2542 ↪ http://www.w3.org/2003/11/swrl#AtomList> ;
2543
2544 ↪ rdf:type <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2545
2546 ↪ <http://www.w3.org/2003/11/swrl#propertyPredicate> :locatedIn ;
2547
2548 ↪ <http://www.w3.org/2003/11/swrl#argument1> :s ;
2549
2550 ↪ <http://www.w3.org/2003/11/swrl#argument2> :c
2551
2552 ↪ ;
2553
2554 ↪ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2555
2556 ↪ rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2557
2558 ↪
2559 ↪ <http://www.w3.org/2003/11/swrl#propertyPredicate> :availableInCountry ;
2560
2561 ↪
2562 ↪ <http://www.w3.org/2003/11/swrl#argument1> :platform ;
2563
2564 ↪
2565 ↪ <http://www.w3.org/2003/11/swrl#argument2> :c
2566
2567 ↪ ] ;
2568
2569 ↪ rdf:rest rdf:nil
2570
2571 ]
2572
2573 ]
2574
2575 ]
2576
2577 ]
2578
2579 ]
2580
2581 ] ;
2582
2583 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2584
2585 rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2586
2587 <http://www.w3.org/2003/11/swrl#classPredicate> :
2588
2589 ↪ FullAccessSubscriber ;
2590
2591 <http://www.w3.org/2003/11/swrl#argument1> :s
2592
2593 ] ;
2594
2595 rdf:rest rdf:nil
2596
2597 ]
2598
2599 ] .
2600
2601 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2602
2603 rdfs:comment "If a review has a score greater than or equal to 7, it's a PositiveReview." ;
2604
2605 rdfs:label "PositiveReviewRule" ;
2606
2607 rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2608
2609 <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2610
2611 rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2612
2613 <http://www.w3.org/2003/11/swrl#classPredicate> :Review
2614
2615 ↪ ;
2616
2617 <http://www.w3.org/2003/11/swrl#argument1> :r
2618
2619 ] ;
2620
2621 rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2622
2623 rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2624
2625 ↪ DatavaluedPropertyAtom> ;
2626
2627 <http://www.w3.org/2003/11/swrl#
2628
2629 ↪ propertyPredicate> :reviewScore ;
2630
2631 <http://www.w3.org/2003/11/swrl#argument1> :
2632
2633 ↪ r ;
2634
2635 <http://www.w3.org/2003/11/swrl#argument2> :
2636
2637 ↪ score
2638
2639 ] ;
2640
2641 rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
2642
2643 ↪ AtomList> ;
2644
2645 rdf:first [ rdf:type <http://www.w3.org
2646
2647 ↪ /2003/11/swrl#BuiltinAtom> ;
2648
2649 <http://www.w3.org/2003/11/swrl#
2650
2651 ↪ builtin> <http://www.w3.org/2003/11/swrl#greaterThanOrEqual> ;
2652
2653 <http://www.w3.org/2003/11/swrl#
2654
2655 ↪ arguments> [ rdf:type rdf:List ;
2656
2657
2658

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2589         ↪         rdf:first :score ;
2590         ↪         rdf:rest [ rdf:type rdf:List ;
2591         ↪             rdf:first 9 ;
2592         ↪             rdf:rest rdf:nil
2593         ↪         ]
2594         ↪     ]
2595         ↪         rdf:rest rdf:nil
2596         ↪     ]
2597     ]
2598 ] ;
2599 <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2600     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2601         ↪ <http://www.w3.org/2003/11/swrl#classPredicate> :
2602         ↪ PositiveReview ;
2603         ↪ <http://www.w3.org/2003/11/swrl#argument1> :r
2604         ↪ ] ;
2605         ↪ rdf:rest rdf:nil
2606     ] .
2607 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2608     rdfs:comment "Natural language: If a subscriber watched a series, and that series has a season, then the
2609         ↪ subscriber has also engaged with that season." ;
2610     rdfs:label "allSeasonsRule" ;
2611     rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2612     <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2613         ↪ rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2614             ↪ <http://www.w3.org/2003/11/swrl#classPredicate> :
2615             ↪ Subscriber ;
2616             ↪ <http://www.w3.org/2003/11/swrl#argument1> :s
2617             ↪ ] ;
2618             ↪ rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2619                 ↪ rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2620                     ↪ IndividualPropertyAtom> ;
2621                     ↪ <http://www.w3.org/2003/11/swrl#
2622                     ↪ propertyPredicate> :watched ;
2623                     ↪ <http://www.w3.org/2003/11/swrl#argument1> :
2624                     ↪ s ;
2625                     ↪ <http://www.w3.org/2003/11/swrl#argument2> :
2626                     ↪ series
2627                     ↪ ] ;
2628                     ↪ rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
2629                         ↪ AtomList> ;
2630                         ↪ rdf:first [ rdf:type <http://www.w3.org/
2631                             ↪ /2003/11/swrl#ClassAtom> ;
2632                             ↪ <http://www.w3.org/2003/11/swrl#
2633                             ↪ classPredicate> :Series ;
2634                             ↪ <http://www.w3.org/2003/11/swrl#
2635                             ↪ argument1> :series
2636                             ↪ ] ;
2637                             ↪ rdf:rest [ rdf:type <http://www.w3.org/
2638                                 ↪ /2003/11/swrl#AtomList> ;
2639                                 ↪ rdf:first [ rdf:type <http://www.
2640                                     ↪ w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2641                                     ↪ <http://www.w3.org/
2642                                     ↪ /2003/11/swrl#propertyPredicate> :hasSeason ;
2643                                     ↪ <http://www.w3.org/
2644                                     ↪ /2003/11/swrl#argument1> :series ;
2645                                     ↪ <http://www.w3.org/
2646                                     ↪ /2003/11/swrl#argument2> :season
2647                                     ↪ ] ;
2648                                     ↪ rdf:rest rdf:nil
2649                                 ↪ ]
2650                             ↪ ]
2651                         ↪ ]
2652                     ↪ ] ;
2653                 ↪ ] ;
2654             ↪ ] ;
2655         ↪ ] ;
2656         ↪ <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2657             ↪ rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2658                 ↪ IndividualPropertyAtom> ;
2659                 ↪ <http://www.w3.org/2003/11/swrl#propertyPredicate> :

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2642         ↪ watched ;
2643                                     <http://www.w3.org/2003/11/swrl#argument1> :s ;
2644                                     <http://www.w3.org/2003/11/swrl#argument2> :season
2645                                     ] ;
2646         rdf:rest rdf:nil
2647     ] .
2648
2649 [ <http://swrl.stanford.edu/ontologies/3.3/swrla.owl#isRuleEnabled> "true"^^xsd:boolean ;
2650   rdfs:comment "If A and B are friends and both watched the same artwork, A co-watched it with B." ;
2651   rdfs:label "cowatchedRule" ;
2652   rdf:type <http://www.w3.org/2003/11/swrl#Imp> ;
2653   <http://www.w3.org/2003/11/swrl#body> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2654     rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#ClassAtom> ;
2655       <http://www.w3.org/2003/11/swrl#classPredicate> :
2656         ↪ Subscriber ;
2657       <http://www.w3.org/2003/11/swrl#argument1> :a
2658     ] ;
2659     rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2660       rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2661         ↪ ClassAtom> ;
2662         ↪ classPredicate> :Subscriber ;
2663         ↪ b
2664       ] ;
2665       rdf:rest [ rdf:type <http://www.w3.org/2003/11/swrl#
2666         ↪ AtomList> ;
2667         ↪ rdf:first [ rdf:type <http://www.w3.org
2668           ↪ /2003/11/swrl#IndividualPropertyAtom> ;
2669           ↪ <http://www.w3.org/2003/11/swrl#
2670             ↪ propertyPredicate> :friendOf ;
2671             ↪ <http://www.w3.org/2003/11/swrl#
2672             ↪ argument1> :a ;
2673             ↪ <http://www.w3.org/2003/11/swrl#
2674             ↪ argument2> :b
2675           ] ;
2676           rdf:rest [ rdf:type <http://www.w3.org
2677             ↪ /2003/11/swrl#AtomList> ;
2678             ↪ rdf:first [ rdf:type <http://www.
2679               ↪ w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2680               ↪ <http://www.w3.org
2681               ↪ /2003/11/swrl#propertyPredicate> :watched ;
2682               ↪ <http://www.w3.org
2683               ↪ /2003/11/swrl#argument1> :a ;
2684               ↪ <http://www.w3.org
2685               ↪ /2003/11/swrl#argument2> :x
2686             ] ;
2687             rdf:rest [ rdf:type <http://www.w3
2688               ↪ .org/2003/11/swrl#AtomList> ;
2689               ↪ rdf:first [ rdf:type <
2690                 ↪ http://www.w3.org/2003/11/swrl#IndividualPropertyAtom> ;
2691                 ↪ <http://www
2692                 ↪ .w3.org/2003/11/swrl#propertyPredicate> :watched ;
2693                 ↪ <http://www
2694                 ↪ .w3.org/2003/11/swrl#argument1> :b ;
2695                 ↪ <http://www
2696                 ↪ .w3.org/2003/11/swrl#argument2> :x
2697               ] ;
2698               rdf:rest rdf:nil
2699             ]
2700           ]
2701         ]
2702       ] ;
2703     <http://www.w3.org/2003/11/swrl#head> [ rdf:type <http://www.w3.org/2003/11/swrl#AtomList> ;
2704       rdf:first [ rdf:type <http://www.w3.org/2003/11/swrl#
2705         ↪ IndividualPropertyAtom> ;
2706         ↪ <http://www.w3.org/2003/11/swrl#propertyPredicate> :
2707         ↪ coWatched ;
2708         ↪ <http://www.w3.org/2003/11/swrl#argument1> :a ;
2709         ↪ <http://www.w3.org/2003/11/swrl#argument2> :b
2710       ] ;
2711       rdf:rest rdf:nil
2712     ]
2713 ] .

```

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2697

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